Characterizing Limb Tremor as a Multi-Degree of Freedom (MDOF) System Involving Springs and Dampers

Ray Ramadhar, Clarence Tang, Hyungkyu Lee, Brayan Ramos Vicente

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Abstract

Upper-limb tremor was modeled as a lumped-parameter multi-degree of freedom (MDOF) system to study tremor propagation through the shoulder, elbow/forearm, and wrist. The arm was represented using seven rotational degrees of freedom (q): shoulder flexion, shoulder adduction, shoulder internal rotation, elbow flexion, forearm pronation, wrist flexion, and wrist ulnar deviation. Published stiffness, damping, and inertia matrices were used in the equation of motion, and the system was converted to state-space form for time and frequency domain analysis. A sinusoidal tremor torque of $\tau(t) = \frac{1}{3}\sin(8\pi t)$ was applied one degree of freedom at a time, representing a 4 Hz tremor input within the 4–12 Hz or occasionally widened 2-12 Hz tremor band. The model system was stable and the natural frequencies were 0.67, 1.08, 1.60, 1.88, 3.05, 4.64, and 8.57 Hz, with damping ratios ranging from 0.147 to 0.839.

The first six resonant frequencies ranged from 0.65 to 2.96 Hz, while the seventh mode did not produce real resonant frequency because of its high damping ratio. Time-domain and frequency-response results showed that the largest tremor responses occurred in the distal limb degrees of freedom (DOFs). At 4 Hz, forearm pronation reached 0.440 rad, or about 25.2°, wrist flexion reached 0.310 rad, or about 17.8°, and wrist ulnar deviation reached 0.095 rad, or about 5.4°. These results corresponded to FRF gains of 1.341, 0.944, and 0.282 rad/N·m, respectively. A passive suppression concept was then modeled as a rigidly attached weighted wristband by adding a device inertia matrix to the original arm inertia matrix. For the 4 Hz tremor case, the added inertia reduced the forearm pronation response from about 25.2° to about 18.9° and reduced the wrist flexion response from about 17.8° to about 0.44°. However, the wrist ulnar deviation response increased from about 5.4° to about 7.4°. These results indicate that an inertia-based wristband can strongly reduce wrist flexion tremor and partially reduce forearm pronation

tremor, but the current design is not uniformly effective across all distal degrees of freedom and may worsen wrist ulnar deviation.

Introduction

Problem Statement

There is still a lot unknown about Upper Limb Tremor and how the condition varies from person to person. Though there have been massive leaps in suppression devices, continued research on the topic will give insight into the limb mechanics and tremor source and propagation. This will allow for further iteration of such devices and more affordable options for those in need. Using knowledge of MDOF systems, the limb mechanics of Upper Limb Tremor were split into 7 DOF's: 3 were for the shoulder, 2 were for the elbow, and 2 were for the wrist. Using researched impedance matrices, and sinusoidal tremor inputs in different DOF's, the system was put in state space form and analyzed in the time and frequency domains. Through this analysis, the determination of the system characteristics ($\omega_n, \zeta, \omega_r$) and mode shapes, tremor propagation and coupling could be analyzed. From this, a theoretical suppression device was designed based on the baseline results [1].

Background

Upper Limb tremor is typically a result of Parkinson's disease or Essential Tremor. It is often characterized as a tremor of varying magnitude during daily activities and actions such as eating, buttoning a shirt and more. This tremor can occur over a broad age range but typically occurs in middle aged and older individuals. This condition is accompanied by a mild degeneration of the cerebellum and can also be genetically inherited. There are medications and suppression devices available but there is still much progress to be made, to serve a broader

population (Tremor | National Institute of Neurological Disorders and Stroke). Tremor is defined by its origin, its propagation, and its distribution throughout the entire arm. This tremor can originate from the shoulder, elbow, and wrist joints, and can occur over varying amounts of time, and they typically cause shaking in the hands. While information varies from source to source, the tremor band, the range of frequencies of these tremors, is generally from $4 - 12\ Hz$, but can go as low as 2 Hz. The tremors are defined as such: Mild $(1 - 5°)$, Moderate $(5 - 10°)$, Severe $(10° <)$ [2].

Sketch

This schematic demonstrates the 7-DOF breakdown of the arm. The first 3 DOF's are associated with the shoulder, and their angular displacements are correspondingly represented by $q_1, q_2,$ and q_3. These motion displacements are associated shoulder flexion, shoulder adduction, shoulder internal rotation, respectively. The 4th (q_4) and 5th (q_5) DOF's are representative of elbow flexion, and forearm pronation. Lastly, the 6th (q_6) and 7th (q_7) DOF reflect the angular motion caused by wrist flexion, and wrist ulnar deviation. The chosen posture is also demonstrated in the schematic [1].

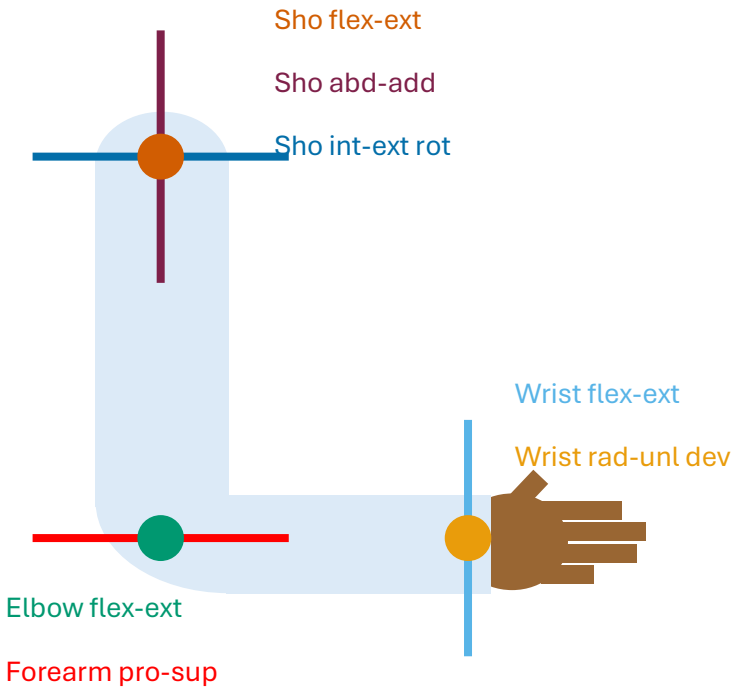


Figure 1: Posture 1 sketch of shoulder, elbow and wrist DOFs

Assumptions

- The arm is modeled as a linear, time-invariant MDOF system.
- The system is treated as a lumped-parameter model using seven rotational DOFs.
- The posture is assumed to stay constant during analysis.
- The I, R, and K matrices are assumed to remain constant.
- Tremor is modeled as a sinusoidal torque input.
- Tremor torque is applied to one DOF at a time to study propagation.
- The same torque amplitude is used for each DOF for comparison.
- Joint motion is assumed to be mostly rotational, not translational.
- Active muscle control, reflexes, and voluntary correction are neglected.
- Coupling between joints is represented by the off-diagonal terms in the matrices.
- The suppression device is modeled as a passive inertial-based device.
- The device acts as if it was rigidly attached to the wrist.

System Modeling

7-DOF Arm Model

Regardless of where the tremor originates, the coupling between the limbs, which is mostly a result of the inertial elements, can cause propagation. These input recurring tremor torques can be resolved into sinusoidal torques [1],

$$\tau(t) = Asin(\omega_t t)$$

in each DOF that are of equal magnitude (A) and tremor frequency (ω_t). While the magnitude of torque can vary, the tremor band remains the same. Assuming a unit torque of $A = \frac{1}{3}\ Nm$, and a tremor frequency of 4 Hz, the recurring torque can be written as

$$\tau(t) = \frac{1}{3}sin(8\pi t).$$

Moderate tremors are generally $0.01 - 0.1\ Nm$, while more severe tremors are tremors are $0.1 - 0.5\ Nm$. This input torque as a function of time could then be placed at each DOF, and the angular displacement as a function of time responses could be analyzed to understand propagation. This input torque does not affect the frequency domain FRF or mode shapes, but these responses could prove the amplification and attenuation demonstrated in the time domain responses.

Another crucial factor for analysis is the posture chosen. For this analysis Posture 1, which is standard for many daily activities was chosen. While the damping and stiffness of the arm are mostly independent of posture, different postures can majorly affect the internal

elements of the arm. Since coupling is mostly inertial, this means that posture changes coupling. Coupling generally occurs for DOF's on the same axis, for example, DOF 4 and 7.

The Equation of Motion

The equation of motion for this posture can be defined through angular position, defined by the matrix q. Since there are 7 DOF's angular displacements in joint rotation, q is a 7x1 matrix with components denoted by q_i, for i ranging from 1:7. To account for the stiffness (K), damping (R), and inertial (I) elements for each part of the limb, there is a corresponding 7x7 matrix for each. τ is used to denote the input torques at each joint for each type of joint rotation. Therefore, it is of size 7x1 with elements defined by τ_i for i ranging from 1-7 [1]. The equation of motion can then be defined as

$$I\ddot{q} + R\dot{q} + Kq = \tau.$$

Here, the ith angular displacements and input torque refer to shoulder flexion, shoulder adduction, shoulder internal rotation, elbow flexion, forearm pronation, wrist flexion, and wrist ulnar deviation, for i going from 1-7, respectively. For this analysis, τ is only nonzero for one DOF at a time to accurately simulate tremors.

Using the motion displayed in the sketch above and researched values, a stiffness (K), damping (R), and inertial (I) matrix were created. Since this is a 7 DOF system the matrices were all of size 7x7. The respective matrices from [1] were

$$K = \begin{bmatrix} 10.80 & 2.626 & 0.279 & 2.670 & 0 & 0 & 0 \\ 2.626 & 5.468 & 3.821 & 0 & 0 & 0 & 0 \\ 0.279 & 3.821 & 7.486 & 0 & 0 & 0 & 0 \\ 2.670 & 0 & 0 & 8.670 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0.756 & 0.018 & 0.291 \\ 0 & 0 & 0 & 0 & 0.018 & 0.992 & -0.099 \\ 0 & 0 & 0 & 0 & 0.291 & -0.099 & 2.920 \end{bmatrix} \left[\frac{Nm}{rad}\right],$$

$$R = \begin{bmatrix} 0.756 & 0.184 & 0.020 & 0.187 & 0 & 0 & 0 \\ 0.184 & 0.383 & 0.267 & 0 & 0 & 0 & 0 \\ 0.020 & 0.267 & 0.524 & 0 & 0 & 0 & 0 \\ 0.187 & 0 & 0 & 0.607 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0.021 & 0.001 & 0.008 \\ 0 & 0 & 0 & 0 & 0.001 & 0.028 & -0.003 \\ 0 & 0 & 0 & 0 & 0.008 & -0.003 & 0.082 \end{bmatrix} \left[\frac{Nms}{rad}\right],$$

$$I = \begin{bmatrix} 0.269 & 0 & 0 & 0.076 & 0 & 0 & -0.014 \\ 0 & 0.196 & 0.083 & 0 & -0.002 & 0.009 & 0 \\ 0 & 0.083 & 0.079 & 0 & 0 & 0.011 & 0 \\ 0.076 & 0 & 0 & 0.076 & 0 & 0 & -0.012 \\ 0 & -0.002 & 0 & 0 & 0.002 & 0 & 0 \\ 0 & 0.009 & 0.011 & 0 & 0 & 0.003 & 0 \\ -0.014 & 0 & 0 & -0.012 & 0 & 0 & 0.003 \end{bmatrix} [kgm^2].$$

These matrices demonstrate the principle of tremor propagation that is the main issue with this condition [1]. Diagonal components represent the effects of the joint torques on their respective DOF's, while the non-diagonal components represent the effects of the joint torques on other DOF's. The more non-diagonal elements there are the more coupling in the system there will be. From here, it can be predicted that the inertial elements will cause the most coupling as they have the most non-zero non-diagonal components. This is also expected as stiffness and damping in the arm are mostly determined by muscle stretch.

Analysis

Limitations in Laplace Domain

Typically, the system could be easily defined in the Laplace domain and provide results in both the time and frequency domain. By applying the following impedance transformations,

$$K_{imp}(s) = \frac{K}{s},$$

$$R_{imp}(s) = R,$$

$$I_{imp}(s) = I * s,$$

the impedance matrices for stiffness, damping, and inertial components could be summed elementwise to produce the total impedance matrix Z.

$$Z(s) = K_{imp}(s) + R_{imp}(s) + I_{imp}(s)$$

From here, the admittance could be defined as

$$Y(s) = Z^{-1}(s),$$

And the FRF as

$$H(s) = \frac{Y(s)}{s}.$$

The angular displacement could then be defined by

$$Q(s) = H(s) * T(s),$$

and converted to the time domain using the impulse function. However, due to the way the transfer functions were formed after the inverses were taken, MATLAB struggled to properly display the time responses of the model, with the angular displacement and velocities of the system approaching infinity, which is not physically feasible for a human limb. This behavior is displayed in the following figures. Therefore a more robust methodology, the state space form, was needed to model this system.

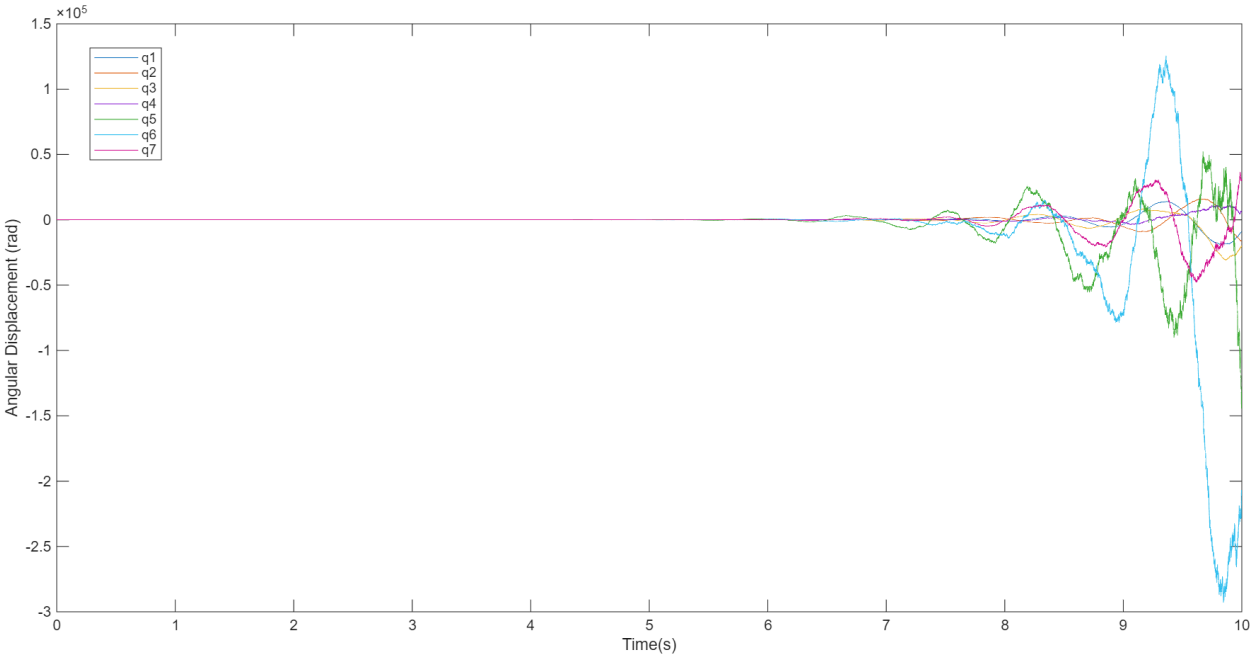


Figure 2: Angular position for each DOF using Laplace method (unstable)

State-Space Conversion

For further analysis the equation of motion was converted to state space form,

$$\dot{x} = Ax + Bu$$

$$y = Cx + Du.$$

The state vector, x, was defined as $\begin{bmatrix} q \\ \dot{q} \end{bmatrix}$ and $u = \tau$. The matrices *A, B, C,* and *D* were defined given that E is the identity matrix, and F is the zero matrix, both of size 7x7. Isolating $\ddot{q}$ yields

$$I\ddot{q} + R\dot{q} + Kq = \tau \rightarrow \ddot{q} = I^{-1}\tau - I^{-1}R\dot{q} - I^{-1}Kq,$$

therefore,

$$A = \begin{bmatrix} F & E \\ -I^{-1}K & -I^{-1}R \end{bmatrix}, \; B = \begin{bmatrix} F \\ I^{-1} \end{bmatrix}, \; C = \begin{bmatrix} E & F \end{bmatrix}, \; D = F$$

This system was defined in MATLAB as shown in the Appendix, using $ss(A, B, C. D)$. The system was then put through the $minreal(\,)$ function in matlab with a tolderance of 10^{-4}, using $sys_{clean} = minreal(sys_{ss}, tol);$. This function can remove certain unobservable or uncontrollable states and can also do pole-zero cancellations. The $balreal(\,)$ function was then used in the line $[sys_{balanced}, \sim] = balreal(sys_{clean});$. This ensured a balanced, potentially reduced order, state-space realization. The combination of the mineral( ) and balreal( ) function ensured feasible plots in the time domain.

System Stability and Characteristics

Given that this system yields a 14th order polynomial for the transfer functions, there are 14 poles for the system. These are 7 pairs of complex poles since the system is underdamped. By analyzing the location of the poles, which were all in the left hand plane as shown in Figure 3 , the system was determined to be stable.

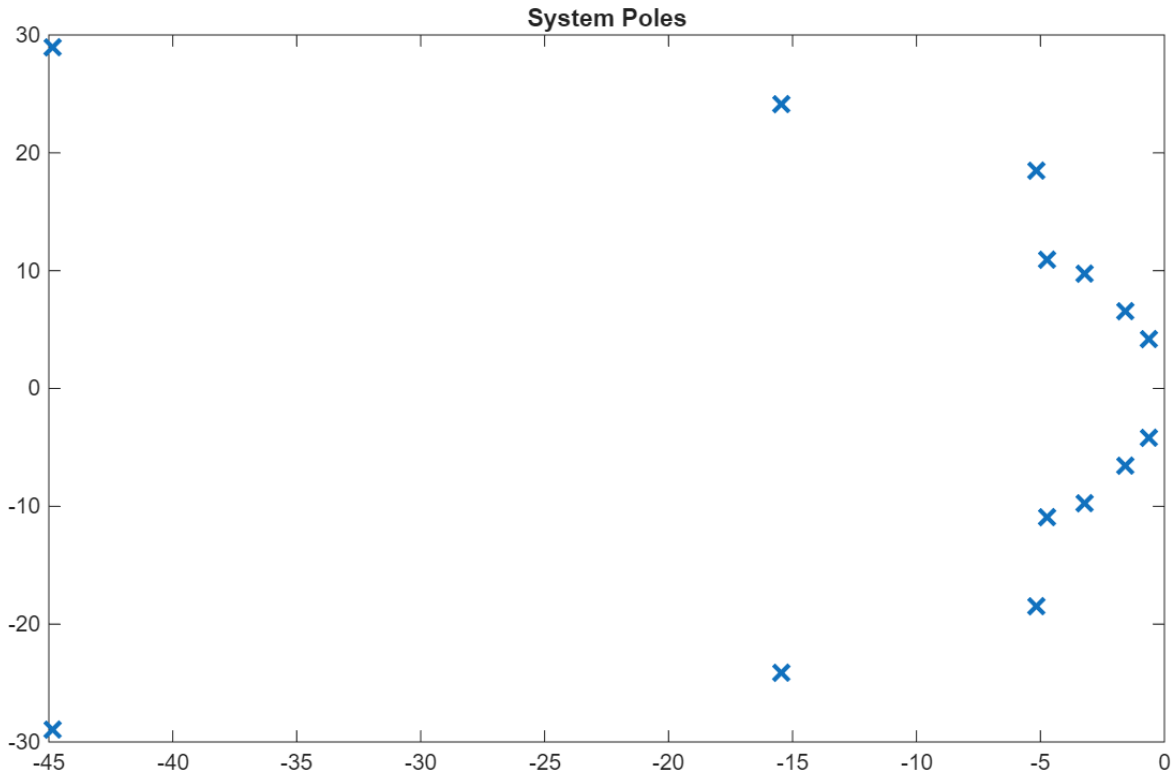


Figure 3: Graph of system poles for stability determination

From here, using the state space form can allow for the determination of the natural frequencies, damping ratio, and consequently the resonant frequencies of the model. The angular displacement for each DOF could also be graphed as a function of time. The magnitude of the frequency response function (FRF), and real and imaginary parts of the FRF could also be plotted. The Mode shapes could then be plotted using the extracted natural frequencies and peak amplitudes.

Implementation of a Suppression Device

To reduce the magnitude of discomfort associated with the tremors, without the need of surgery, passive or active controls can be used. Active controls consist of sensors to measure the vibrations and actuators to counteract the vibrational motion. Therefore, active controls can also

use PID controls to resist the tremor and will require some sort of power sourcing. Passive controls use mechanical elements, such as masses, springs, dampers, to allow for absorbing vibrational energy. For this study, a passive device will be analyzed due to their affordability and simpler operation. These devices must also not affect actions in daily life such as eating, using your phone etc. The comfort for this study will be defined by psychological comfort, sensory comfort and ergonomic comfort.

Results and Simulation

System Characteristics

The natural frequencies and damping ratios were calculated and stored in an array. Since, the model was in state space form, the $damp$ function could be used to extract these quanties easily with the line $[wn_{both}, zeta_{both}] = damp(sys_{balanced});$. This however extracted two duplicate natural frequencies and damping ratios creating a 14x1 array for each rather than a 7x1. The duplicates were removed using $wns = unique(wn_{both});, zetas = unique(zeta_both);$. The resonant frequency of each pole could then be calculated using,

$$\omega_r = \omega_n\sqrt{1 - 2\zeta^2}.$$

The corresponding values for these system characteristics are shown in Table 1

Table 1: System characteristics of the model

Pole Number	$\omega_n(Hz)$	ζ	$\omega_r\ (Hz)$
1	0.67	0.147	0.65
2	1.08	0.235	1.02

3	1.60	0.269	1.51
4	1.88	0.313	1.70
5	3.05	0.398	2.52
6	4.64	0.538	2.96
7	8.57	0.839	-

Time-Domain Response

Methods

After the state space representation was cleaned using the $minreal$ and $balreal$ functions the tremor torque could be implemented. A 7x1 matrix was made to represent the $u = zeros(length(t), 7)$. This set all the torques initially equal to zero. Every individual DOF was assigned the previously determined sinusoidal torque input of $\tau(t) = \sin(8\pi t)$ using $u(:, i) = \frac{1}{3} * sin(8\pi * t)$;. Then system was simulated using $[q, t] = lsim(sys_balanced, u, t)$;, and all angular displacements responses to the origin input torque at each DOF ($i = 1{:}7$) were plotted.

For input tremor torque in DOF 1($\omega_t = 4\ Hz$):

When the input tremor torque was in the positive shoulder flexion DOF, there was minimal movement in all DOF's. The maximum movement occurred at the q_1 (shoulder flexion) and q_4 (elbow flexion), which seemingly move out of phase with each other, and followed by q_7 (wrist ulnar deviation). The transient movement of q_1 initially reaches around 0.009 rad (0.51°) and quickly fall to about 0.003 rad (0.17°) with q_4. These three DOF's maintain their amplitudes through time as the torque remains constant. These tremors still are below what is expected for even a mild tremor, which is around ($1 - 5°$).

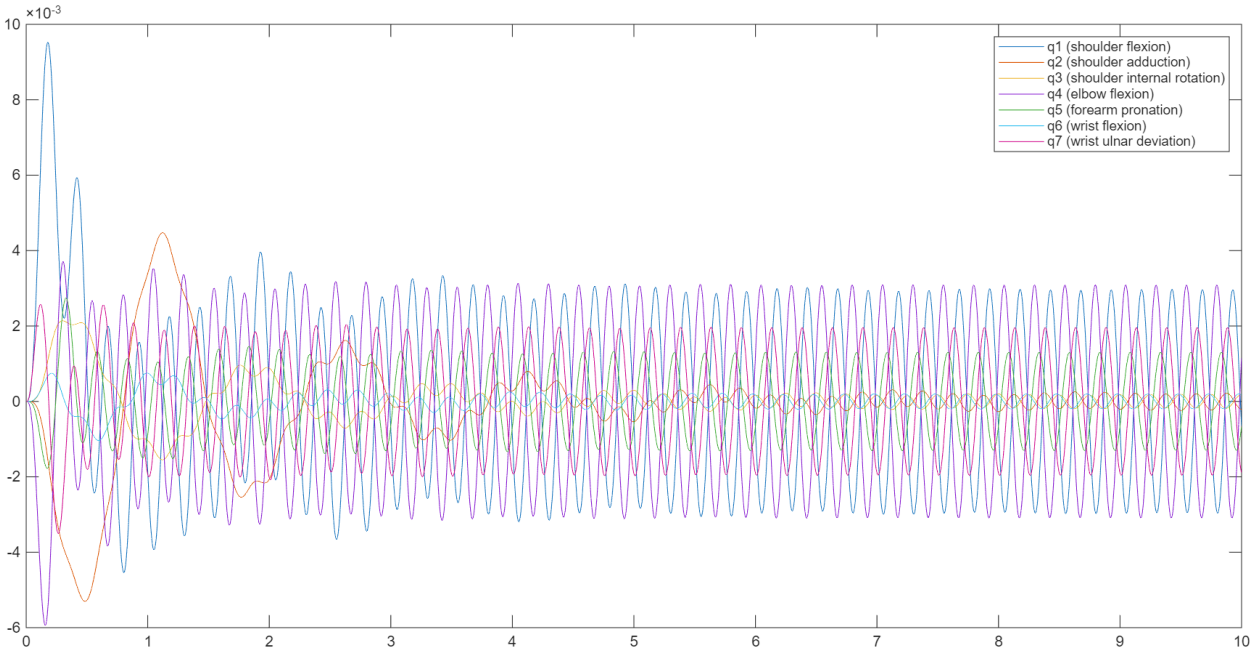


Figure 4: Response to Input Tremor Torque in DOF 1 Over Time

For input tremor torque in DOF 2 ($\omega_t = 4\ Hz$):

When the input tremor torque was in the positive shoulder abduction DOF, there was a slightly higher degree of movement in the wrist. The maximum movement occurred at the q_6 (wrist flexion), q_5 (forearm pronation), and q_2 (shoulder abduction). After a large transient response in the DOF where the torque was active in, with amplitudes reaching 0.019 rad (1.08°). After steady oscillations were met, q_6 maintined an oscillating amplitude of 0.011 rad (0.63°). This again remains below the mild tremor region in the steady state.

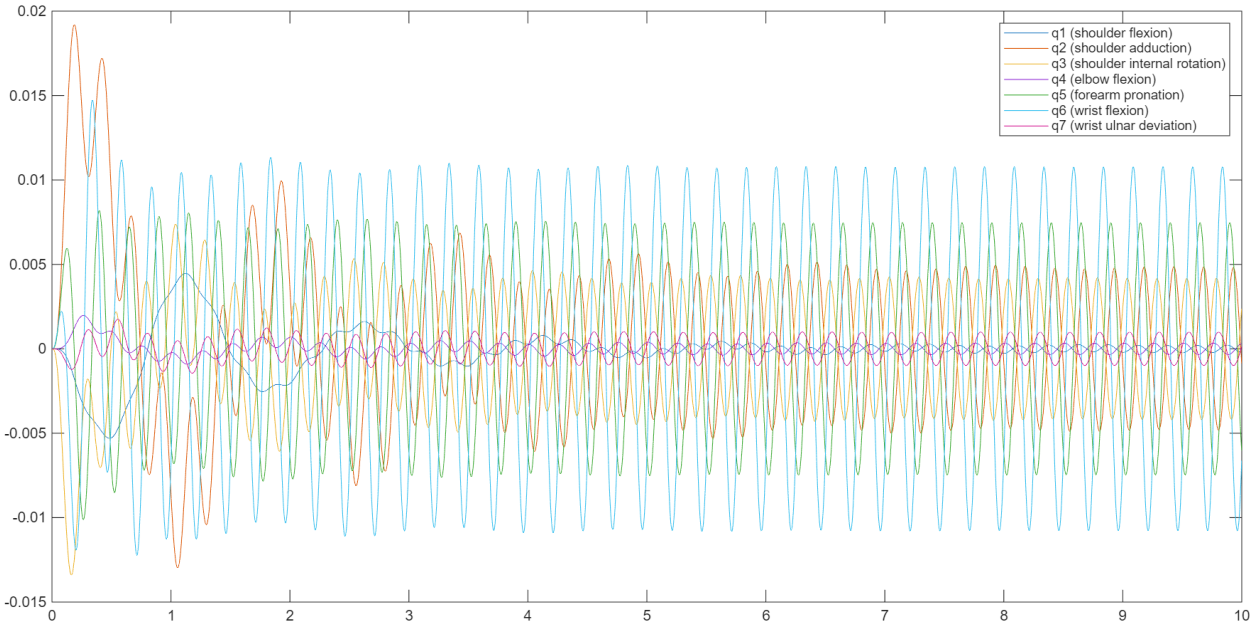


Figure 5: Response to Input Tremor Torque in DOF 2 Over Time

For input tremor torque in DOF 3 ($\omega_t = 4\ Hz$):

When the input tremor torque was in the positive shoulder internal rotation DOF, there was a significant degree of movement in the wrist flexion DOF. After the transient behavior for the shoulder internal rotation settled to about 0.011 rad (0.63°), the wrist flexion DOF osciallated with peak heights of 0.0565 rad (3.32°). This indicates that when the tremor originated in the 3rd DOF (shoulder internal rotation), there was a mild tremor in the wrist flexion DOF. Therefore, tremors seemingly only originating from this particular shoulder DOF have the potential of causing substantial tremors. It should also be noted that q_3 and q_6 are both along the vertical axis.

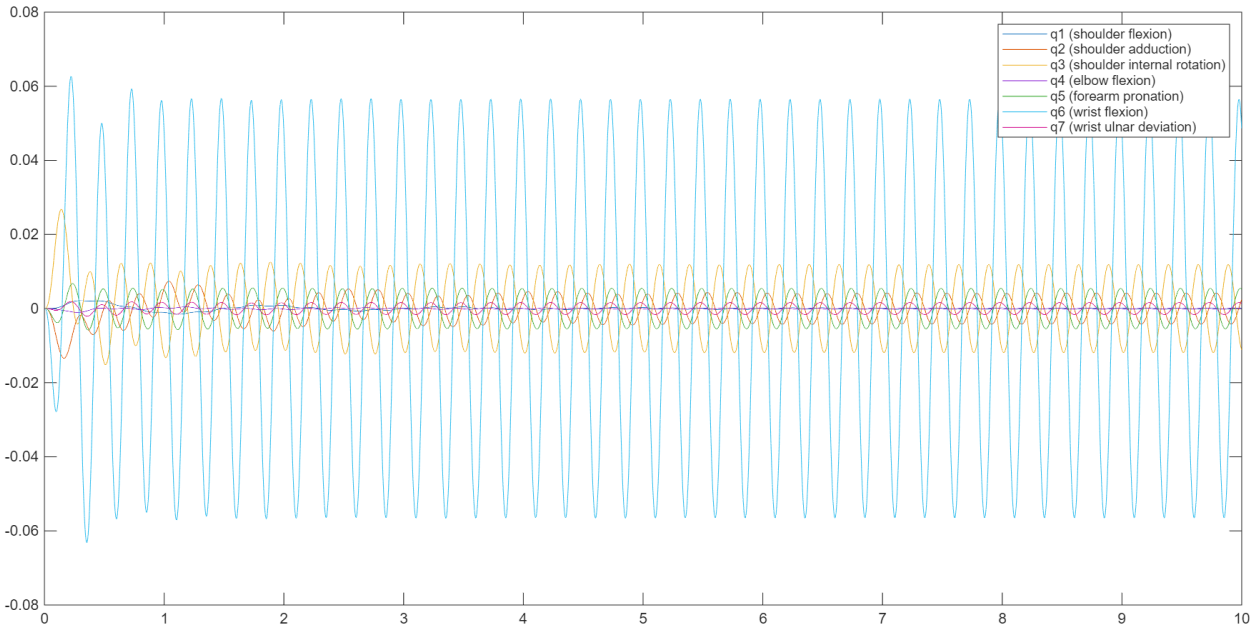


Figure 6: Response to Input Tremor Torque in DOF 3 Over Time

For input tremor torque in DOF 4 ($\omega_t = 4\ Hz$):

When the input tremor torque was in the positive elbow flexion DOF there was little propagation to other DOF's. The maximum angular displacement occurred in the (q_7) wrist ulnar deviation DOF at 0.017 rad (0.97°) and followed by the q_4 (elbow flexion) and q_5 (forearm pronation) DOF's. q_4 and q_7 are along the same axis, the positive z. q_4 and q_5 are also of the same limb region. This reflects little to mild amounts of tremor particularly in the wrist once again.

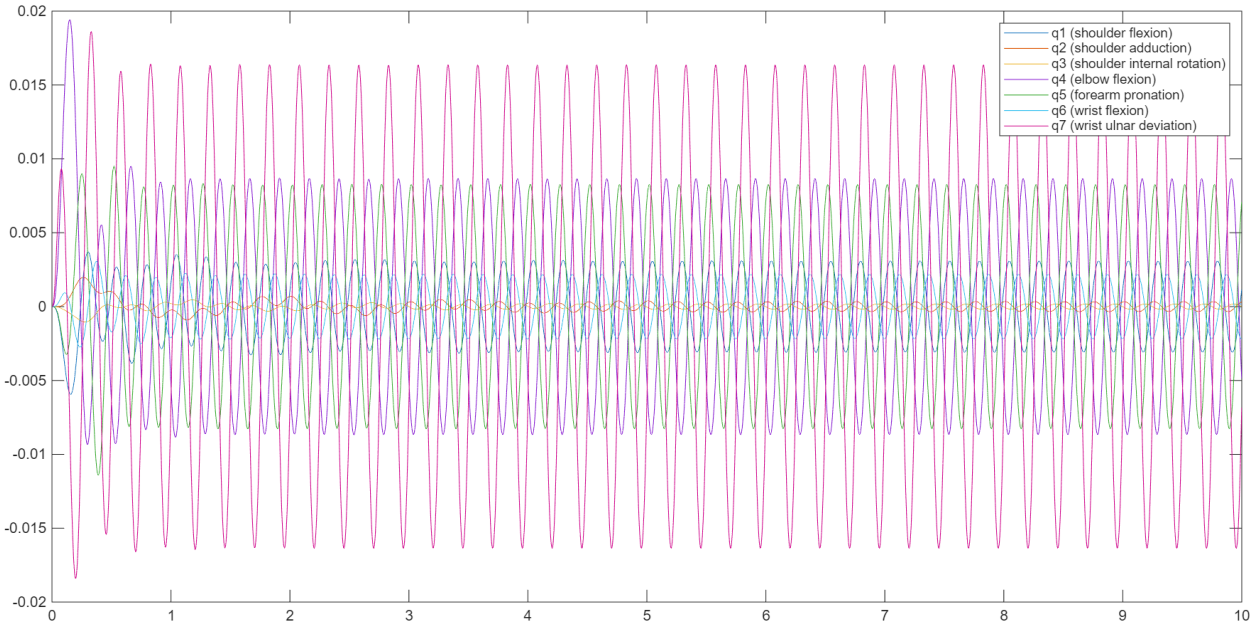


Figure 7: Response to Input Tremor Torque in DOF 4 Over Time

For input tremor torque in DOF 5 ($\omega_t = 4\ Hz$):

When the input tremor torque was in the positive forearm pronation DOF there was substantial angular displacement, the most in any DOF. While many other DOF's had little angular displacement, meaning little tremor propagating in this DOF, q_5, the forearm pronation DOF itself had about 0.440 rad (24.6°) of oscillating movement. This is classified as an extremely severe tremor and could indicate that the input torque amplitude of $\frac{1}{3}Nm$ may have been too large for this system.

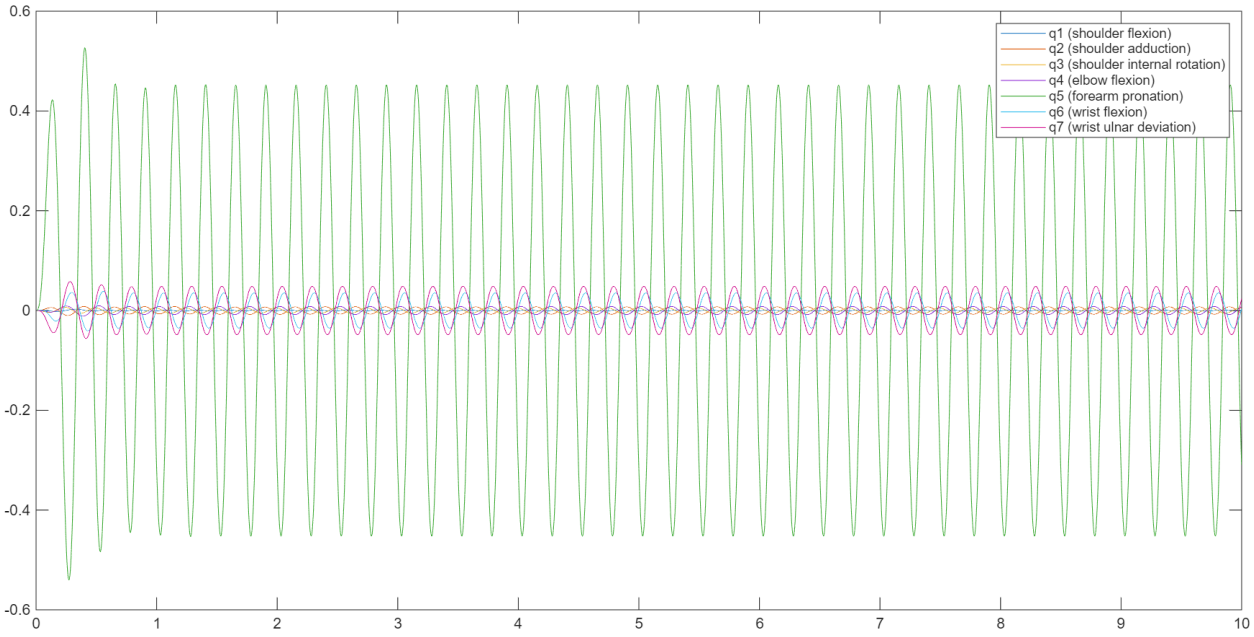


Figure 8: Response to Input Tremor Torque in DOF 5 Over Time

For input tremor torque in DOF 6 ($\omega_t = 4\ Hz$):

When the input tremor torque was in the positive wrist flexion DOF the behavior was like when the tremor torque was in the forearm pronation DOF. Most other DOF's, apart from q_3 (shoulder internal rotation) with around 0.0567 rad (3.4°) experience much angular displacement. It should also be noted that q_6 and q_3 both occur along the vertical axis. Meanwhile, q_6 had an angular displacement that oscillated between ± 0.31 rad ($\pm 17.8°$). This indicated that for this given input sinsuaoidal tremor in this DOF, there is severe tremor ($> 15°$) in the wrist flexion DOF, and mild tremor in the shoulder internal rotation DOF.

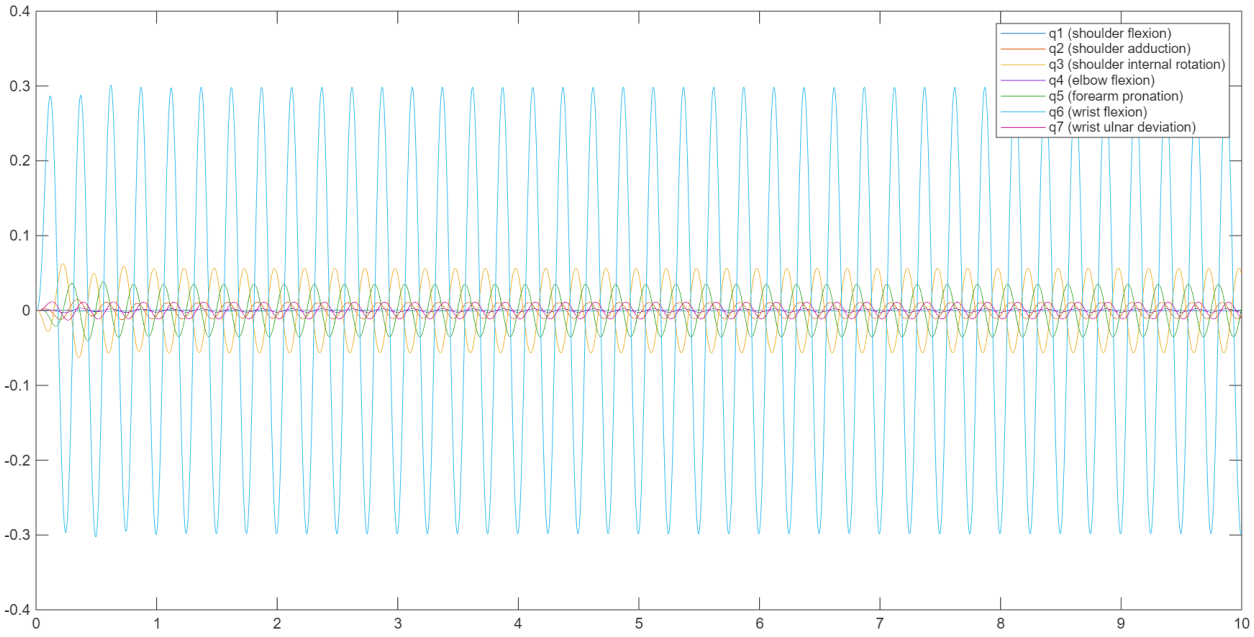


Figure 9: Response to Input Tremor Torque in DOF 6 Over Time

For input tremor torque in DOF 7 ($\omega_t = 4\ Hz$):

When the input tremor torque was in the positive wrist ulnar deviation DOF there was mild tremor given this input tremor torque. The angular displacement in the DOF in which the torque originated, q_7 was around 0.0950 rad (5.84°) and propagated slightly to q_5 (forearm pronation) with a 0.0475 rad (2.72°) displacement. This caused moderate to mild tremor behavior ($5 - 10°$).

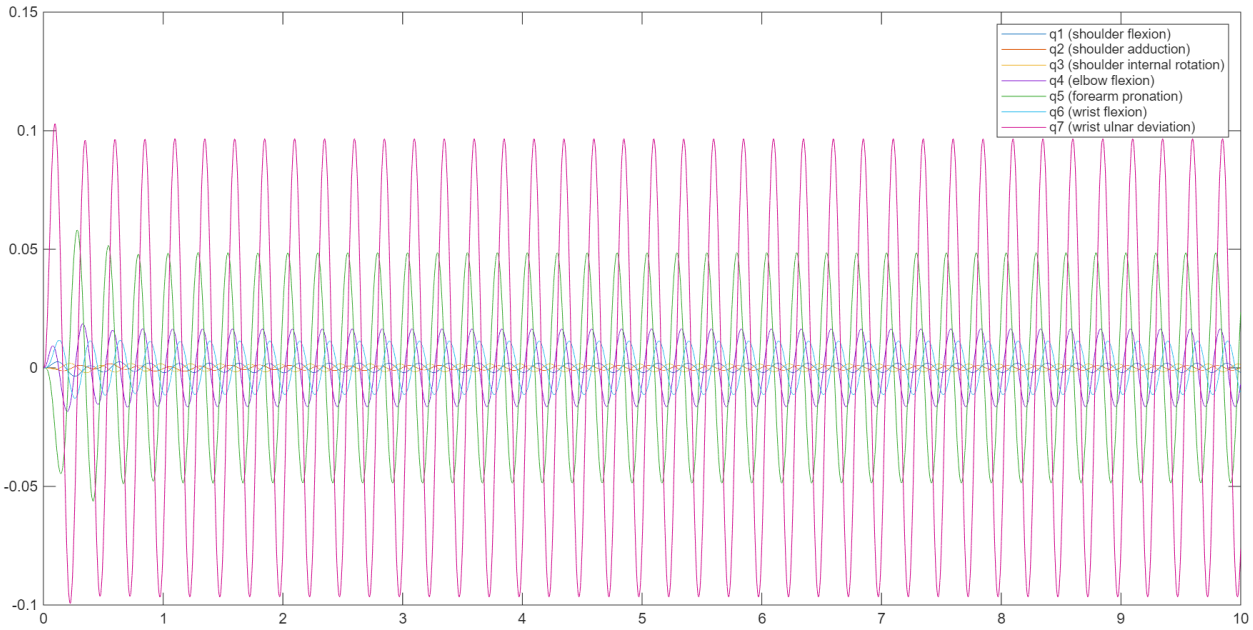


Figure 10: Response to Input Tremor Torque in DOF 7 Over Time

Frequency Response Function (FRF)

Methods

Since the system was already in state space form the frequency response function was extracted using MATLAB's $freqresp$ using the line $freqresp(squeeze(sys_balanced), w)$. After defining the frequency response function, the lines $rowsToPlot = x{:}y$; and $numRows = length(rowsToPlot)$;, were used to define which FRF's were extracted. A for loop, the line $semilogx(w, squeeze(20 * log10(abs((H(i, j, :))))))$; were used to isolate the limb region behavior. To obtain only the FRF response for all shoulder DOF's to all tremor torque inputs, x was set to 1 and y to 3. To obtain only the FRF response for all elbow/forearm DOF's to all tremor torque inputs x was set to 4 and y to 5. To obtain only the FRF response for all wrist DOF's to all tremor torque inputs x was set to 6 and y to 7. The results could then be analyzed to determine the amplification and attenuation for each DOF to all input tremor

locations. Propagation was then analyzed by plotting the reaction of all DOF's to a tremor torque input in each DOF in the appendix.

Shoulder (DOF $1-3$)

For the Shoulder DOFs the peaks generally occurred in the $3-20\ rad/s$ frequency range, though the 7 expected resonant frequency peaks were not all visible. This may be because the natural frequencies were very close. These frequencies were expected given that the calculated system resonant frequencies ranged from $4.1-18.59\frac{rad}{s}$. As shown through the time responses, for the shoulder flexion and shoulder abduction DOF's, the input torques, $\mathrm{T}_{1-7}(s)$, were mostly attenuated. This can be seen through the gains at $\omega = 8\pi\frac{rad}{s}$, the forced frequency, which yielded gains ranging from $0.0005-0.167$. This indicates that the shoulder was not as frequently affected by tremor propagation, and mainly only faced low to mild tremors when the tremor was in the given shoulder DOF. There are also noticeable valleys, also known as antiresonances between the first and second, and the second and third resonant frequencies. These appear to occur at $\omega_{a,s1} = 4.91\frac{rad}{s}$, $\omega_{a,s2} = 5.46\frac{rad}{s}$, $\omega_{a,s3} = 6.41\frac{rad}{s}$, $\omega_{a,s4} = 7.16\frac{rad}{s}$, $\omega_{a,s4} = 7.61\frac{rad}{s}$.

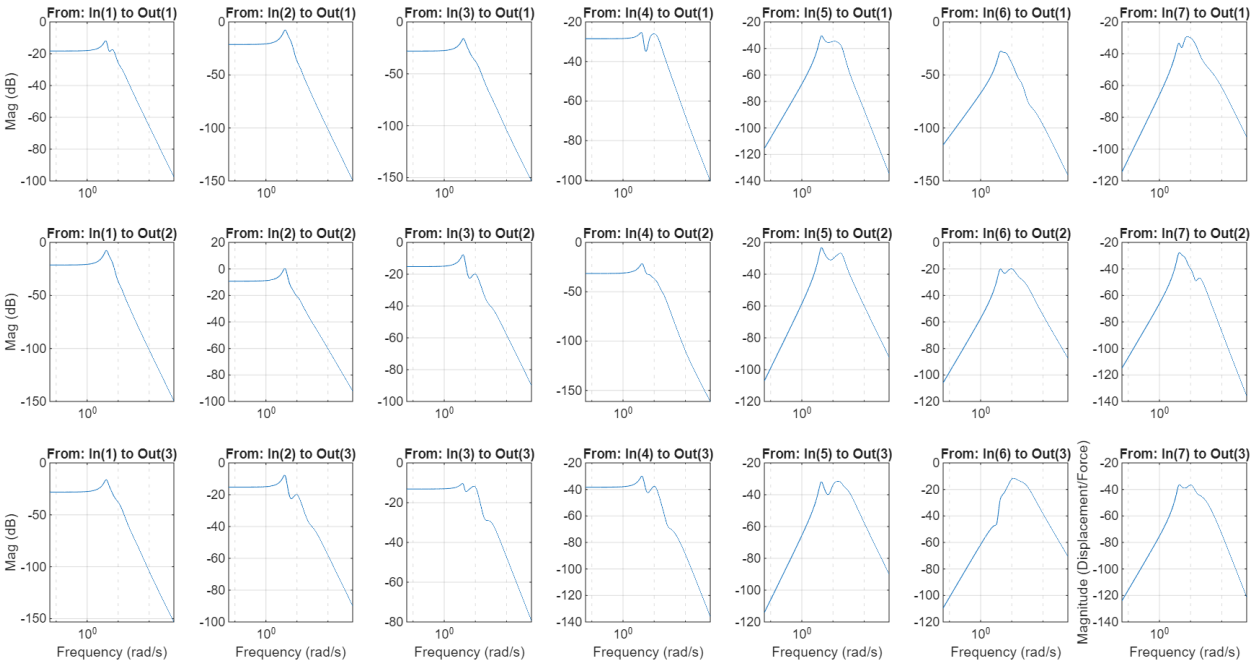


Figure 11: Frequency Responses in the shoulder DOFs to Input Tremor Torques in all DOFs

Elbow/Forearm (DOF $4-5$)

For DOF 4 (elbow flexion) the input torque at different locations with tremor forcing frequency $\omega_t = 8\pi \frac{rad}{s}$ is mostly attenuated for inputs in all DOF's. The gains ranging from $0.004 - 0.05$ explain the lack of major angular displacement due to torque within the DOF and ones in other DOF's in the time domain. While the elbow flexion DOF is at lower risk of even mild tremors the forearm pronation DOF is subject to the most severe tremors with a tremor forcing frequency of $\omega_t = 8\pi \frac{rad}{s}$. At the tremor forcing frequency, where the tremor torque is also in the forearm pronation DOF, the gain is 1.34, indicating that the response $q_5(t)$ is amplified for $\tau_5(t)$. This explains why the angular displacement for this DOF reached heights of 0.44 rad (25.21°).

The forearm pronation DOF maintains a gain of $0.1 - 1.34$ thorughout the $8\pi - 24\pi$ frequency range, indicating a mild to severe tremor is likely with tremor torques originating in the fifth DOF, regardless of the tremor frequency. Propagation of tremors could also be an issue

in the forearm pronation DOF with tremors originating in the wrist ulnar deviation DOF, as shown through the time response for the given amplitude. With a gain of $0.143\,\frac{rad}{Nm}$, mild or even moderate tremors are even possible in the forearm pronation DOF depending on the input torque amplitude for this forcing frequency. There are also anti-resonant dips between the first and second, and the second and third resonant frequencies. These appear to occur at $\omega_{a,e1} = 5.41\,\frac{rad}{s}$, $\omega_{a,e2} = 6.03\,\frac{rad}{s}$, $\omega_{a,e3} = 6.85\,\frac{rad}{s}$, $\omega_{a,e4} = 7.60\,\frac{rad}{s}$.

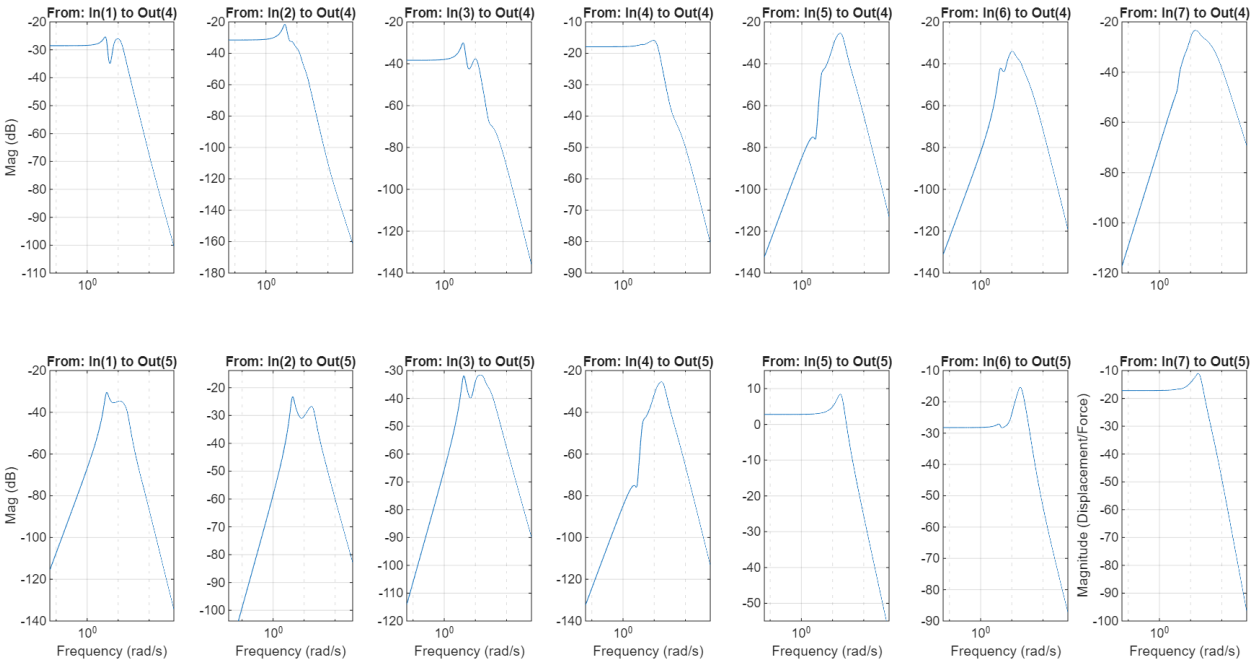


Figure 12: Frequency Responses in Elbow/Forearm DOFs to Input Tremor Torques in all DOFs

Wrist (DOF $6-7$)

For the wrist flexion and wrist ulnar deviation DOF's, all of the tremor frequencies lead to attenuated responses from the degrees of freedom at the wrist. However, while the responses were attenuated, the wrist flexion response $Q_6(s)$ has a gain of $0.944\ rad/Nm$ for $\mathrm{T}_6(s)$. This value being close to 1 means that the displacement in this DOF is severe, since this converts to 0.312 radians or 17.9 degrees. For the wrist ulnar deviation, $Q_7(s)$ has a gain of 0.282, which converts to 0.093 rads or 5.3 degrees. This means that tremors with the tremor frequency of

$8\pi\ rad/s$ cause angular displacement q_7 would cause moderate displacements. The anti-resonant dips appear to occur at $\omega_{a,w1} = 5.15\frac{rad}{s},\ \omega_{a,w2} = 5.28\frac{rad}{s}, \omega_{a,w3} = 5.59\frac{rad}{s}, \omega_{a,w4} = 6.09\frac{rad}{s}$, and $\omega_{a,w3} = 6.60\frac{rad}{s}$.

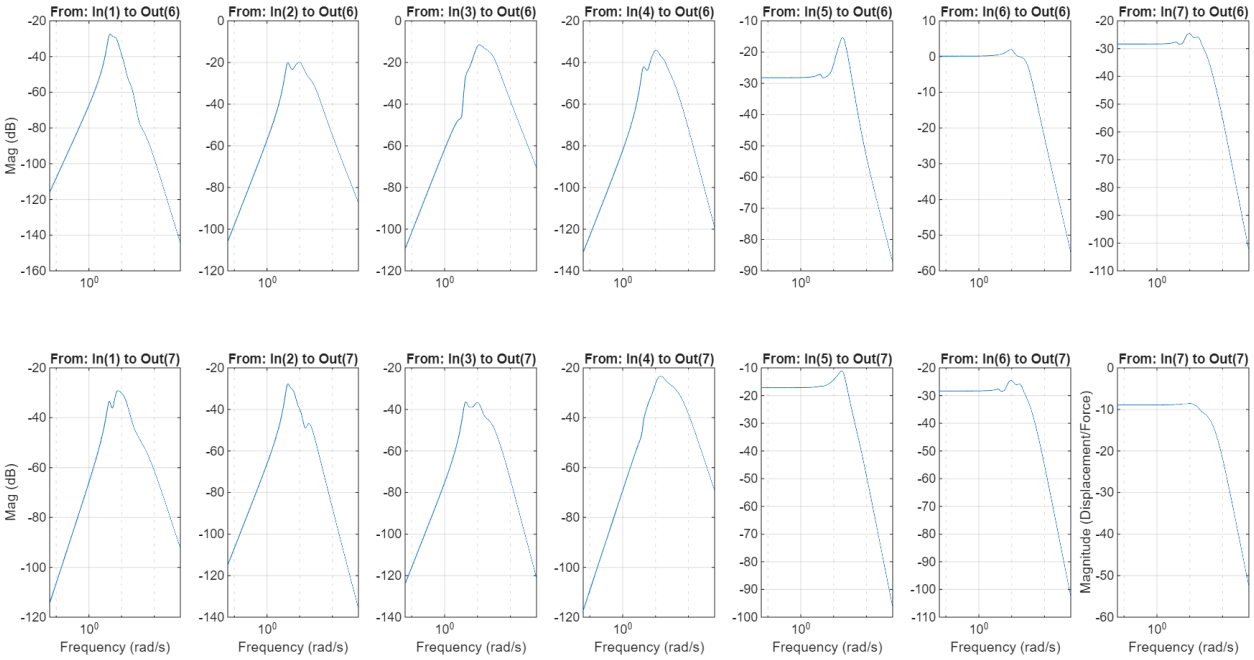


Figure 13: Frequency Responses in Wrist DOFs to Input Tremor Torques at all DOFs

Magnitude Comparison and Tremor Band

To see the gains more clearly the magnitudes of all DOF's with respect to an input torques in each DOF was plotted below. Again, while input torques in the DOFs 1-4 show little gains for almost all DOFs, input torques in DOFs 5-7 show the clear spikes in gain in and around the $2 - 5\ Hz$ low tremor frequency part of the tremor band.

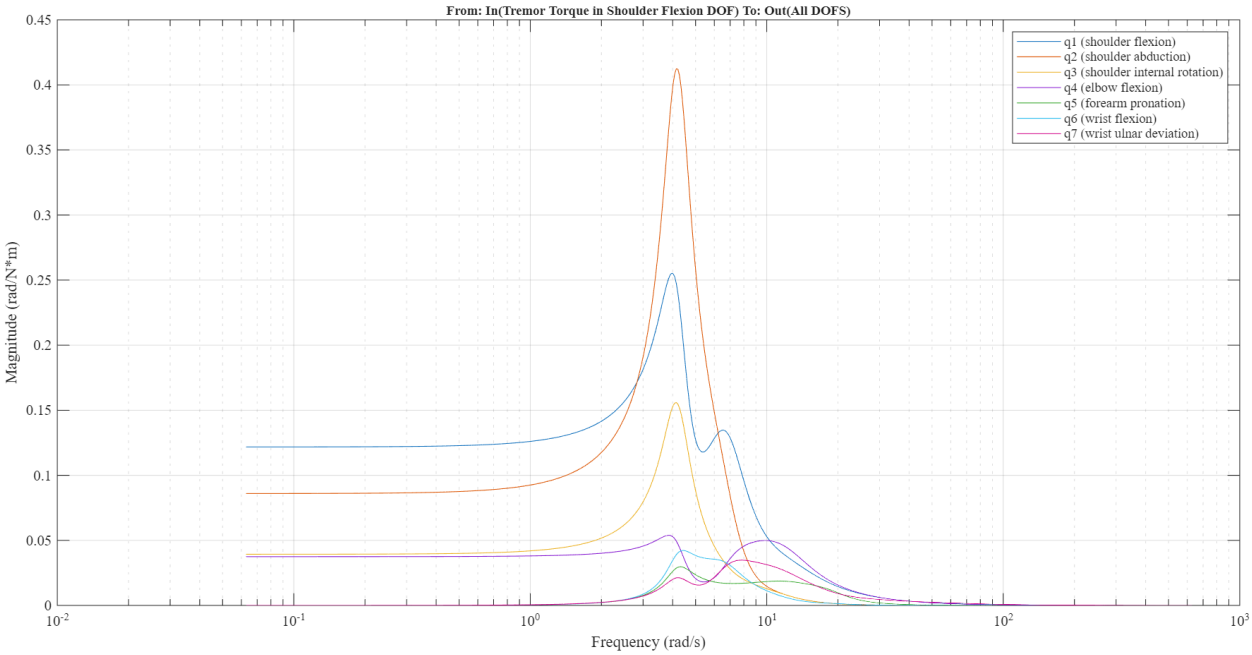


Figure 14: Magnitude of Each DOFs Response to Input Tremor Torque in DOF 1

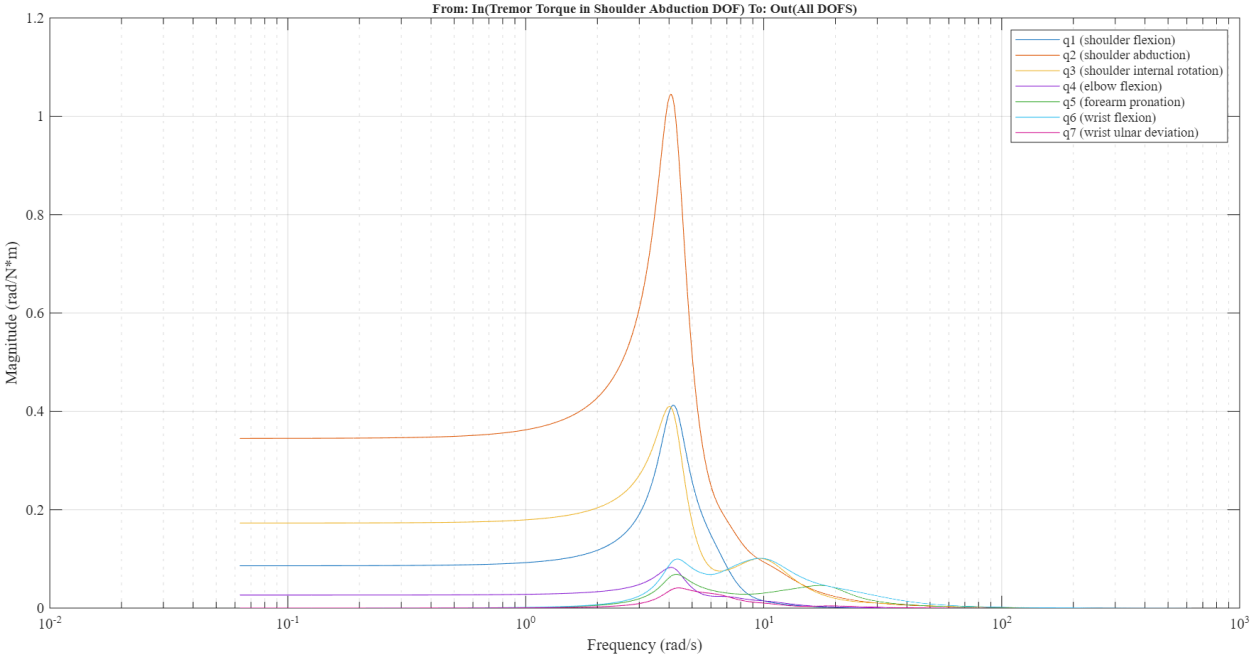


Figure 15: Magnitude of Each DOFs Response to Input Tremor Torque in DOF 2

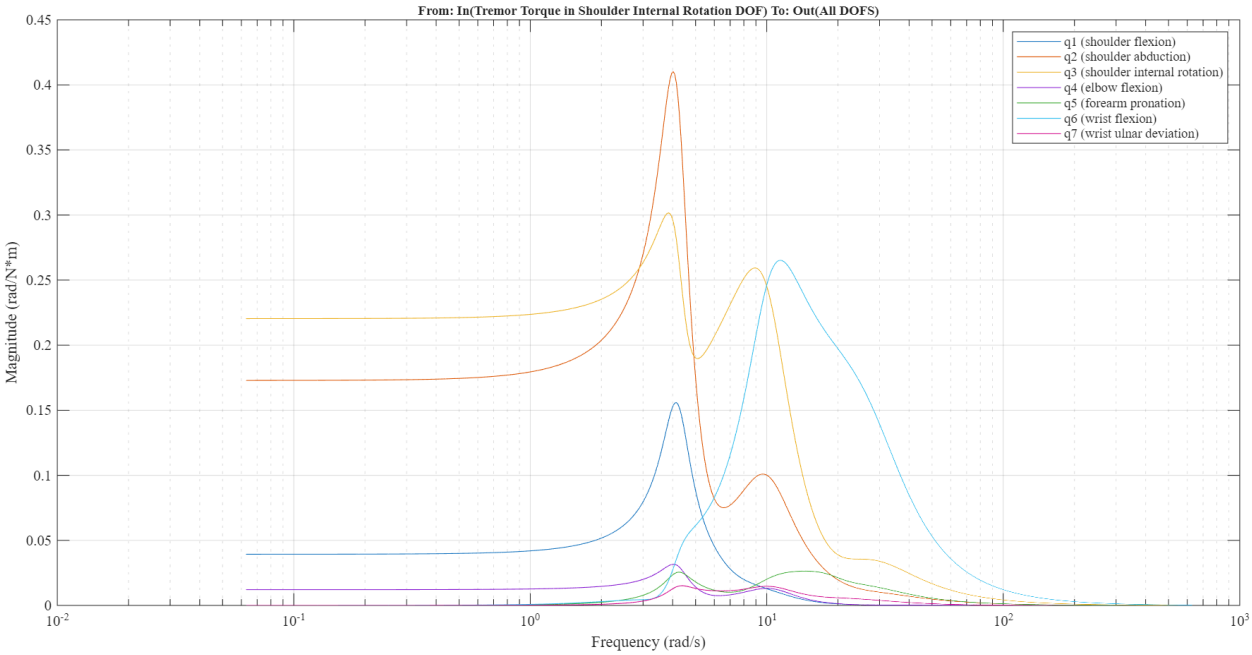


Figure 16: Magnitude of Each DOFs Response to Input Tremor Torque in DOF 3

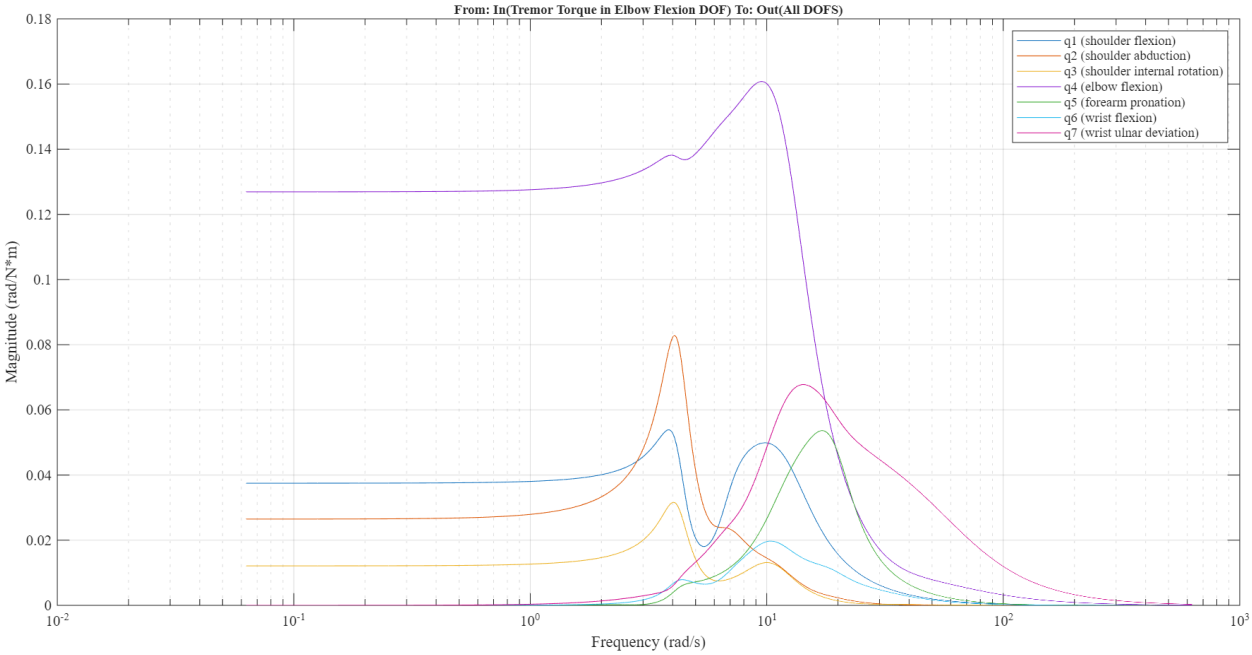


Figure 17: Magnitude of Each DOFs Response to Input Tremor Torque in DOF 4

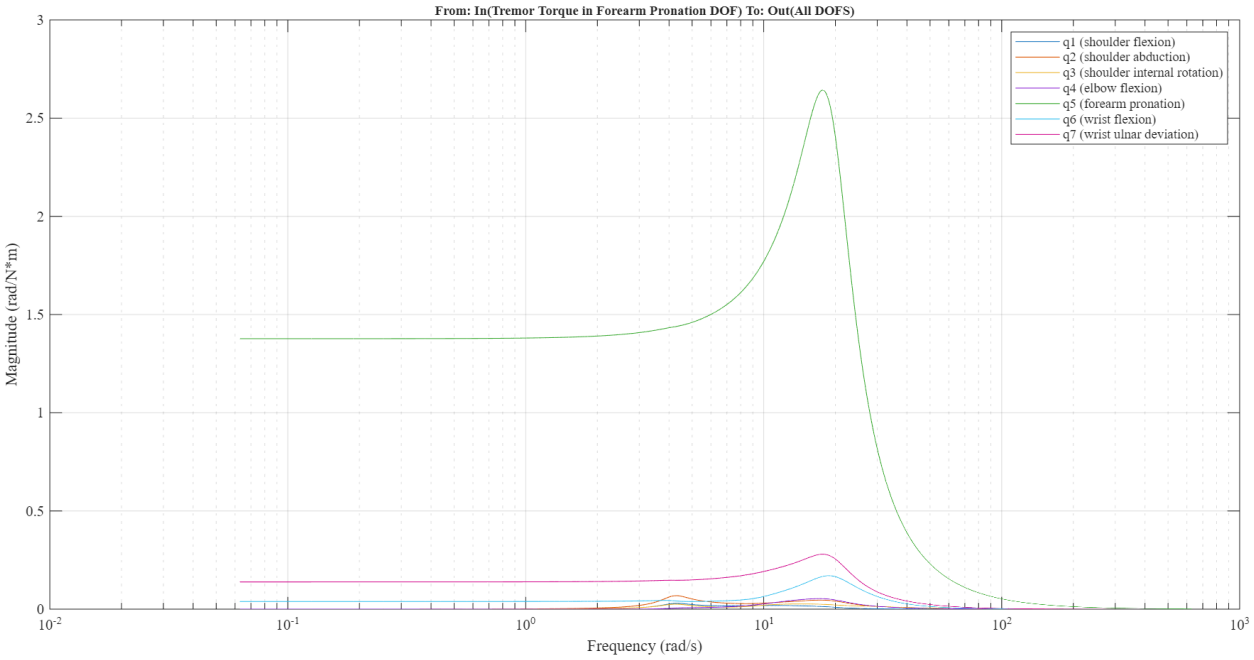


Figure 18: Magnitude of Each DOFs Response to Input Tremor Torque in DOF 5

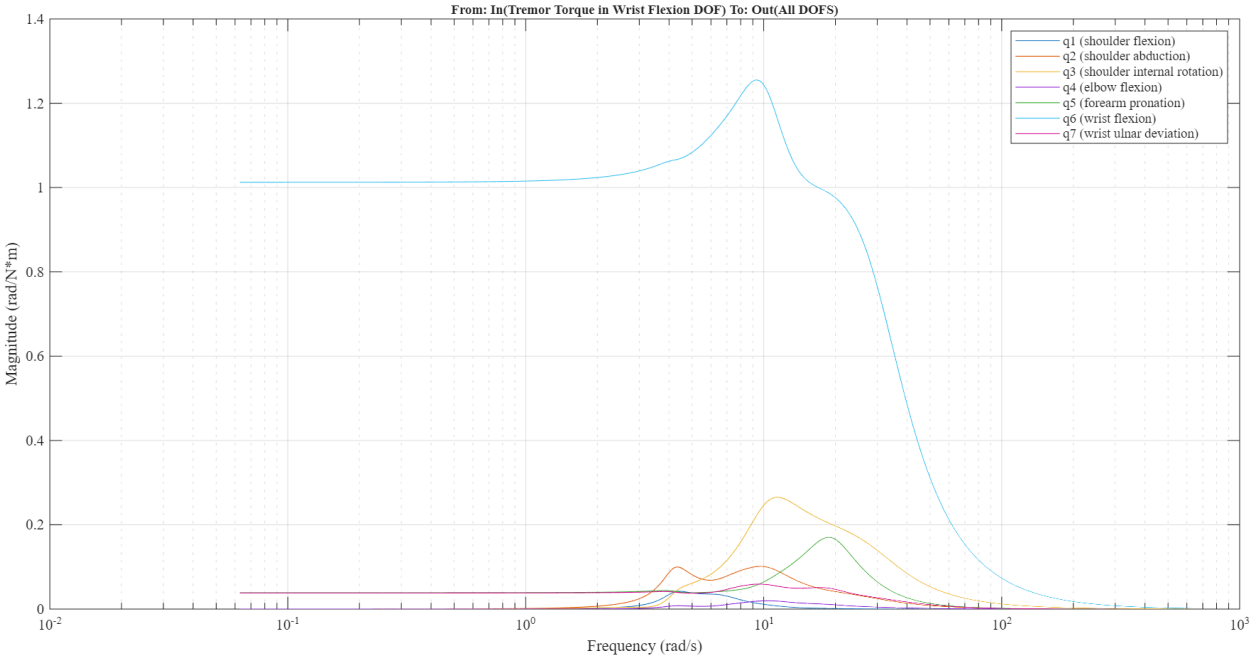


Figure 19: Magnitude of Each DOFs Response to Input Tremor Torque in DOF 6

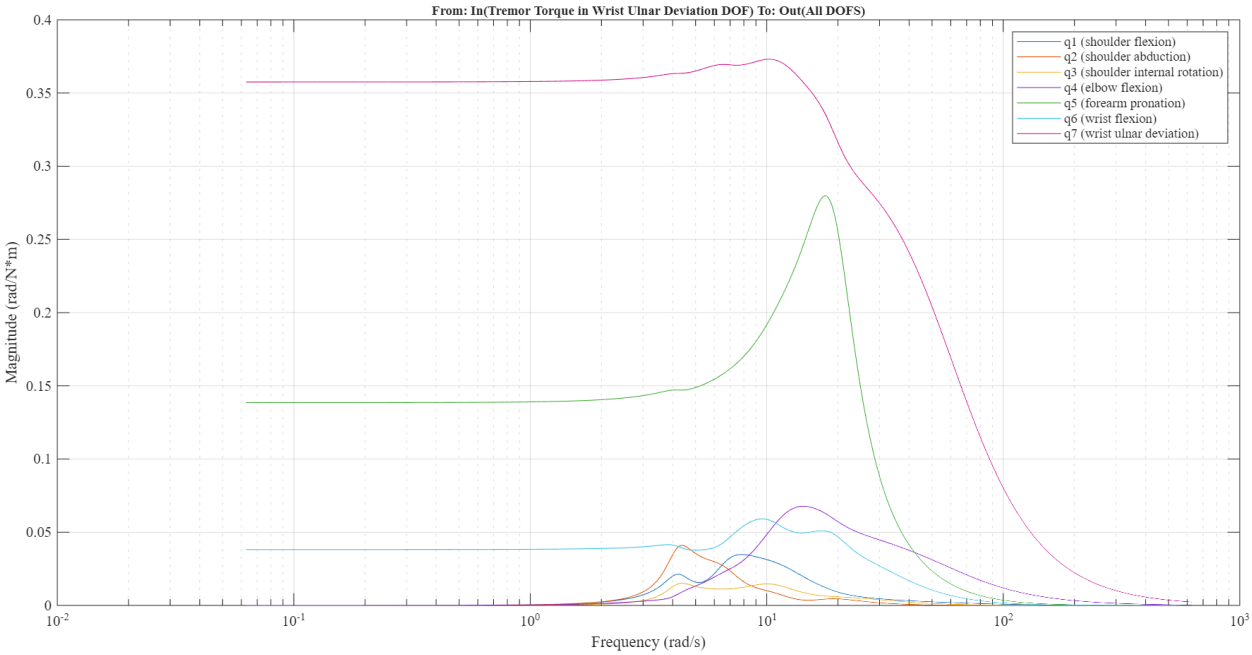


Figure 20: Magnitude of Each DOFs Response to Input Tremor Torque in DOF 7

Mode Shapes

For the mode shapes that were normalized by the maximum value in the specified mode vector, the behaviors at each of the resonant frequencies can be seen. These mode shapes go from top to bottom for clarity. The lower resonant frequencies, such as Modes 1-4, are well outside the tremor band of 4-12 Hz or 2-12 Hz in some studies. This means that they are unlikely to occur in real life, with tremors generally not occurring at such low frequencies. Therefore, the shoulder and elbow region will likely not be the displacements maximized by the tremor torque. Modes 5-7, occurring in a frequency range of $3.05 - 8.57\ Hz$, which is well within the tremor band region, will be much more prominent in individuals with tremor. At these modes forearm pronation, and wrist flexion and ulnar deviation are maximized from Modes 5-7, respectively. This can be proven through video documentation where movement generally occurs in the forearm pronation DOF, and wrist DOF's. This can also be seen more clearly in the non-normalized mode shapes where DOF 5-7 have relatively high displacements at higher modes.

There are also a significant amount of nodes for DOF 1-4 in the modes that have more overlap with the tremor band (right 3).

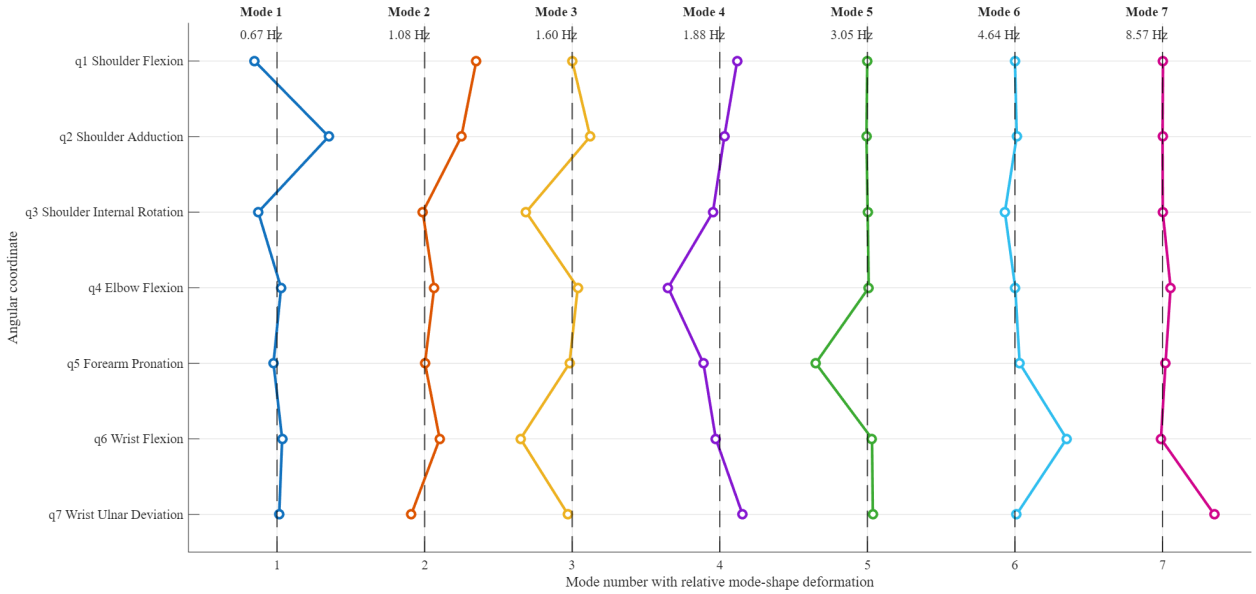


Figure 21: Normalized Mode Shapes of the System

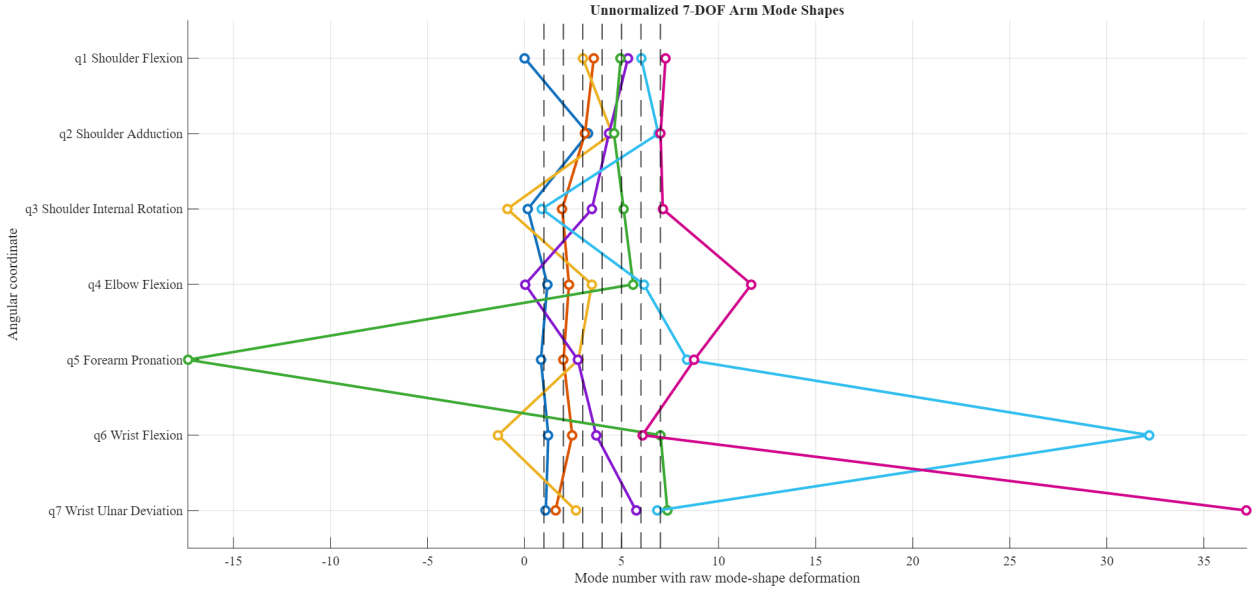


Figure 22: Unnormalized Mode Shapes of the System

Implementation of Tremor Suppression

In order to simulate the tremor suppression device, a weighted wristband was used in hopes of decreasing the most significant angular displacements in the forearm pronation, wrist flexion, and wrist ulnar deviation DOFs. In order to simulate this, a new matrix was created named I_{device} [3]. It was assumed that the wristband was rigidly attached so it did not create an 8th DOF. Using the equations

$$I_{device_{i,j}} = m_r L_i L_j$$

for $i, j = 1{:}7$ and $i = j$ the diagonal entries could be calculated. Then using

$$I_{device_{i,j}} = m_r L_i L_j \cos(\theta_i - \theta_j)$$

for $i, j = 1{:}7$ and $i \neq j$ the non-diagonal entries could be calculated. Only the DOFs rotating in the same axis for this given posture had off diagonal components [3]. Using the estimated parameters $m_w = mass\ of\ wristband = 0.26\ kg$, $L_{UA} = Length\ of\ upper\ arm = 0.3\ m$, $L_F = Length\ of\ forearm = 0.25\ m$, $r_w = Length\ from\ wrist\ to\ wristband = 0.05\ m$ and $R_P = Radius\ of\ Pronation = 0.04\ m$ the equations the matrix was

$$I_{device} = \begin{bmatrix} 0.0468 & 0 & 0 & 0 & 0 & 0 & 0.004446 \\ 0 & 0.0468 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0.0468 & 0 & 0 & 0.004446 & 0 \\ 0 & 0 & 0 & 0.0234 & 0 & 0 & 0.0039 \\ 0 & 0 & 0 & 0 & 0.000416 & 0 & 0 \\ 0 & 0 & 0.004446 & 0 & 0 & 0.0234 & 0 \\ 0.004446 & 0 & 0 & 0.0039 & 0 & 0 & 0.00065 \end{bmatrix},$$

$$I_{total} = I + I_{device}.$$

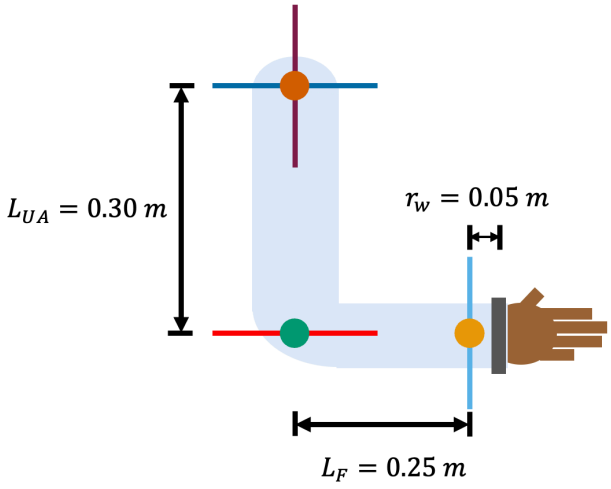


Figure 23: Sketch of Arm with Tremor Suppression Device

This decreased the gain in some cases, or caused the magnitude plot to shift leftward, or caused a small increase or no change, as shown in the plots below for tremor torques in all DOFs. For the FRFs of interest, the (5,5), (6,6), (7,7) magnitude responses, where the tremor was the most significant, the behavior varied significantly. For a tremor with frequency 4 Hz in DOF 5, the gain of the DOF 5 dropped to about $1\frac{rad}{Nm}$, reducing the 25.21° forearm pronation angular displacement to 18.9°. For a tremor with frequency 4 Hz in DOF 6, the gain of the DOF 6 dropped to about $0.0231\frac{rad}{Nm}$, reducing the 17.9° wrist flexion angular displacement to 0.44°. For a tremor with frequency 4 Hz in DOF 7, the gain of the DOF 7 actually increased to about $0.39\frac{rad}{Nm}$, changing the original 5.32° wrist ulnar deviation angular displacement to 7.37°. Therefore, sever tremors of DOF 5 and 6 became less severe and very insignificant, respectively, and the moderate tremors of DOF 7 stayed in the moderate region while increasing slightly.

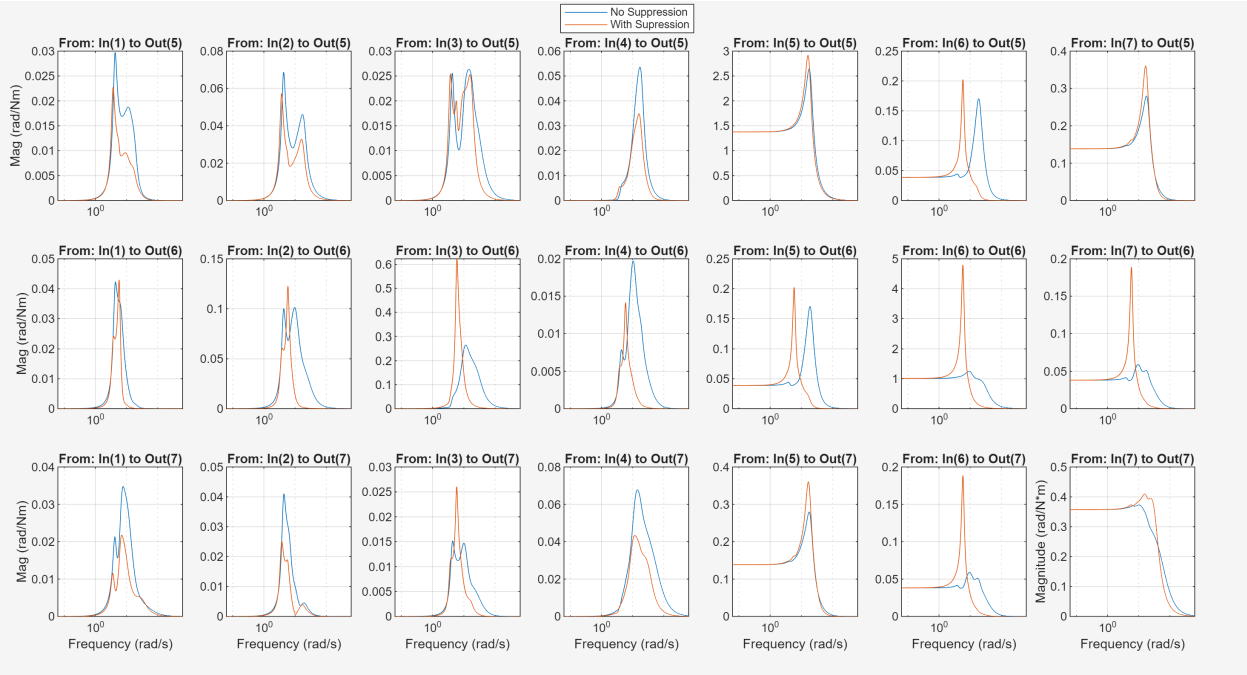


Figure 24: Magnitudes of DOF 5,6,7 to Tremor Torques in all DOFs With/Without Suppression

Discussion

Time-Frequency Domain Connections

For the assumed amplitude and forcing frequency of the tremor torque, $\tau_{1-7}(t) = \frac{1}{3}\sin(8\pi t)$, the relationship between the frequency and time domain responses were clear. For the DOF's demonstrating anywhere from mild to severe tremor given this input torque in a specified DOF, the gains were extracted from the FRF responses using

$$Gain = G = 10^{dB/20},$$

where dB was representative of the magnitude at the forcing frequency, $\omega = 4\,Hz \approx 25.13\frac{rad}{s}$. By multiplying this calculated gain by the torque amplitude (1/3),

$$G * \frac{1}{3} = Amplitude,$$

the peak amplitude in the time domain response for that given DOF could be extracted. These values were compared to the actual peaks in the time domain response to demonstrate the overlap below.

Table 2: Time and Frequency Domain Connections

Location of Tremor Torque (DOF)	Angular displacement of interest (q)	Gain Calculated from FRF at forcing frequency	Peak Amplitude from Time Domain (rad)	Gain * Torque amplitude (rad)	Tremor Type
3 (SIR)	q_6	0.170	0.0565	0.0561	Mild
5 (FP)	q_5	1.341	0.440	0.442	Severe
5 (FP)	q_7	0.143	0.0469	0.0471	Mild
6 (WF)	q_3	0.171	0.0567	0.0564	Mild
6 (WF)	q_6	0.944	0.310	0.312	Severe
7 (WUD)	q_5	0.145	0.0475	0.0477	Mild
7 (WUD)	q_7	0.282	0.0950	0.0930	Moderate

Something also of notice is the reciprocity in the frequency and time Domain. For the frequency domain, all rows $i = 1{:}\,7$ and columns $j = 1{:}\,7$, the FRF of $H(i,j)$ was the same as the the FRF for $H(j,i)$. This affects the response in the time domain as well, as the $q_n(t)$ response to $\tau_m(t)$ at the forcing frequency is the same as the $q_m(t)$ response to $\tau_n(t)$ at the same forcing frequency, for all $n, m = 1{:}\,7$.

The wrist flexion and forearm pronation were the DOF's exhibiting the highest amount of tremor. While the particular angle in radians can drastically change given the torque amplitude the gains calculated reflect the behavior of the DOF with respect to an input torque in the specified DOF. Within their own DOF, the gains were near or above 1 for DOF 5 and 6. Consequently, any input torque applied to these specific DOFs results in a high-sensitivity response, where the output angular displacement meets or exceeds a 1:1 ratio relative to the input.

Tremor Band Analysis and Couplings

The range of frequencies for tremor torques is typically $8\pi - 24\pi \frac{rad}{s}$ $(2 - 12\ Hz)$. While the analysis of the time domain only displays the responses of the angular displacement of each DOF's, the behavior to other experimental forcing frequencies, such as 16π and 24π, can be predicted through the FRF responses. As the frequency across the FRF matrix increased, for inputs at all locations, the magnitude of all of the DOFs' FRF response declined, indicating attenuation. This indicates that while mild-severe tremors can still occur at higher frequency tremor torques, the input torque signal is most greatly amplified for forcing torque frequencies are at the lower end of the tremor frequency range $(2 - 5\ Hz)$. However, at higher forcing frequencies, DOF's with moderate to severe tremors at lower frequencies, such as DOF 5 and 6, still experience mild to moderate tremors with gains ranging from $0.2 - 1$ in the $5 - 12\ Hz$ range.

Another factor that noticeable across the time and frequency domain was the tremor propagation for DOF's in the same axis of rotation, as shown through the time domain with DOF

5 and 7 and 3 and 6. This represents the coupling within the upper limb and originates from the off-diagonal components in the inertial, stiffness, and damping matrices. These mostly occurred in the inertial matrix, signifying that DOF coupling, and therefore tremor propagation, was mostly caused by the inertia attributed to each DOF. Coupling is the prime reason why many of those affected have an intense tremor. Each individual DOF tremor may be declared as mild to moderate, while the combination of all of the local tremor movements may make it significantly harder to complete daily activities than it would have been with the angular displacement of a singular DOF [1].

The given posture therefore has a major impact on the coupling in the system, and therefore the propagation. This is because posture directly changes the inertial matrix due to different mass distributions. The chosen posture of the study, which can encompass mostly resting behavior with little grip strength (using a phone, operating a TV remote etc.), will therefore reflect different natural and resonant frequencies in the upper limbs than a different position. For tighter gripping, like with eating, the impedance matrices would change significantly, as muscle tensing would reflect changes in stiffness and damping matrices, and clenching would also change the mass distribution, hence affecting the inertial matrix. Tremor may also propagate in a different fashion though the relationship of coupling between same-axis movement will likely remain [1].

However, the amplification of these the input torques may be larger, especially if the change in posture reflects new resonant frequencies close to that of the inputted tremor forcing frequencies. The same can be said for attenuation, as the movement of resonant frequencies may cause resonant frequencies and FRF magnitude peaks to occur far outside of the range of the tremor band. For this given posture, many FRF magnitude plots of the various DOF's to the

range of input torque DOFs, had a peak in gain that were within the expected resonant frequency range ($0.67 - 2.96\ Hz$). Therefore, this analysis demonstrated one of the worst case forcing frequency scenarios, with the forcing frequency of $\omega_t = 4\ Hz$, being near this range. This indicates that for this given standard posture, individuals with a lower freuqnecy tremor will experience the worst tremors [4].

This is especially an issue considering that this is one of the most common resting postures, and because most tremors tend to propagate away from the trunk of the body according to studies [3] This means that proximal DOFs, such as the ones located in the shoulder are much less likely to be the origin of the tremor torques. Meanwhile, the DOFs with the most tremor, which are the distal DOFs including forearm pronation, wrist flexion, and wrist ulnar deviation, are the most common origin or the tremor torque. When the torque was in these distal DOFs the most severe tremors and propagation were caused, as shown in the time domain response for input torque in DOFs 5-7 and the right three columns of the FRF responses.

This was also assuming that the torque inputs in all DOFs were constant, which further validates the much more erratic movement of the distal parts of the upper limb. This can be attributed to the inertial matrix which has a clear decrease in value ($0.26 - 0.003$) along the diagonal from proximal to distal, shoulder to wrist. This shifted the peaks gain and magnitudes of the distal features to frequencies that were further right, and more in the tremor band [5]. This can be understood through the equation

$$\omega_n = \sqrt{\frac{k}{I}},$$

showing that lower inertia means higher natural and resonant frequencies. They had peaks that occurred more near the higher system resonant frequencies ($2.52 - 2.96 \frac{rad}{s}$). Meanwhile, the proximal shoulder and even elbow maintained peak gains and magnitudes near the first resonant frequencies ($4.10 - 10.69 \frac{rad}{s}$), with higher inertial values bringing their peak magnitudes mostly outside of the tremor band. This demonstrates the need for assistive devices to help a wide range of individuals in a costly and effective manner, so they can live comfortably.

Impact of Suppression Device using Passive Control

The implementation of device increased the inertia and generally shifted the locations of the magnitude peaks leftward, but still had many flaws. Based on the previously defined relationship with inertia and natural frequency, increasing the inertia in the wrist should shift the magnitude plots leftward, and had the potential to make the peak magnitudes both decrease or increase. For DOF 6, the peaks were well below the lower frequency range of $2 - 5\ Hz$, meaning that lower tremor frequencies, such as the experimented 4 Hz would have little to no effect on wrist flexion, across input DOFs. However, the magnitude peaks of DOF 5 only shifted slightly leftward, decreasing the gain but maintaining severe tremors for inputs in the same DOF. For DOF 7, the wristband caused little to no leftward shift and actually made the angular displacements larger for tremor torque inputs at this frequency in this DOF.

Therefore, the wristbands inertia would likely reduce or potentially eliminate the chances of moderate-severe tremors in wrist flexion DOF, but would have little to no effect on DOF 5 and 7. Adding an additional mass to increase the inertia of the forearm could help significantly to push the peak magnitude gains of DOF 5 below the tremor band could also be helpful. Adding more muscle stiffness for the wrist ulnar deviation could push the magnitude peaks ahead of the

tremor band. However, to implement and simulate both of these strategies diagonal and non-diagonal couplings of the matrices would need to be calculated. Adding in dampers, while complicating these designs, would help to lessen the quality factor and decrease the impact of the new peak magnitude gains. Including the very strong springs would move the peak magnitude gains outside of the far ahead of the tremor band but would also make the device very uncomfortable, and require a lot of exertion from the individuals to even move voluntarily. This would also happen if the mass of the wristband was increased, except the magnitude gains would be way below the tremor band.

Limitations

The limitations to this study mainly concern the choice of the torque amplitude and forcing frequency. While the FRF is unaffected by this and only relies on the impedance matrices, which were put through sensitivity analysis, the time domain directly simulates the expected behavior associated with the torque amplitude and frequency. All of the time domain results and estimations in angular displacement with respect to gain are all dependent on the chosen torque amplitude of $\frac{1}{3} Nm$ and tremor frequency of 4 Hz. The torque value was chosen based off what is expected for a severe tremor torque ($0.1 - 0.5\ Nm$), while the frequency of 4 Hz was used to easily view the different displacements overlayed on each other. For different tremor frequencies and torque amplitudes, the behavior can instead be predicted by the FRF's based off gains. This made the time domain responses more for visualization of a particular scenario. The inertia added to the DOFs by the device could also cause issues for voluntary movement. The bracelet was also considered to be rigidly mounts on the wrist, to avoid another DOF, however, this could also cause some issues with accuracy, but likely not general trends.

Conclusion

The 7-DOF upper-limb tremor model showed that tremor propagation is strongly affected by coupling between degrees of freedom, especially through the inertial matrix. The state-space model produced stable, underdamped dynamics with natural frequencies from 0.57 to 2.96 Hz and damping ratios from 0.147 to 0.839. For the assumed tremor input torque of $\frac{1}{3}\sin(8\pi t)$ N·m at 4 Hz, the shoulder and elbow-flexion DOFs were mostly attenuated, while the distal DOFs showed much larger responses. The most severe predicted tremors occurred in forearm pronation and wrist flexion, where the angular displacements reached approximately 0.440 rad, or 25.2°, and 0.310 rad, or 17.9°, respectively. Wrist ulnar deviation produced a moderate response of about 0.095 rad, or 5.3°. These results were consistent with the frequency-response gains, where forearm pronation had the largest self-response gain of $1.341\ rad/N \cdot m$, followed by wrist flexion at $0.944\ rad/N \cdot m$.

The mode-shape analysis also supported the observation that distal limb motion dominates tremor behavior. Modes 5–7 were associated with forearm pronation, wrist flexion, and wrist ulnar deviation, and their natural frequencies overlapped the broader tremor-frequency range discussed previously. This suggests that tremors originating in or propagating to the forearm and wrist are more likely to produce significant shaking than tremors isolated to the shoulder or elbow. The chosen passive suppression concept was an inertia-based weighted wristband. In the model, the wristband was assumed to be rigidly attached to the limb and was represented by adding a device inertia matrix to the original inertia matrix. This design produced mixed results at the 4 Hz tremor input. Forearm pronation decreased from about 25.2° to about 18.9°, so the response was reduced but still remained large. Wrist flexion showed the strongest

improvement, decreasing from about 17.8° to about 0.44°. In contrast, wrist ulnar deviation increased from about 5.4° to about 7.4°, meaning that the wristband may worsen motion in that DOF. Overall, the weighted wristband is most effective for wrist flexion, partially effective for forearm pronation, and ineffective for wrist ulnar deviation in the current model.

Future work should tune the added inertia, include damping, and investigate a combined inertia–damping or inertia–stiffness design so that the device reduces tremor without shifting or increasing motion in another clinically relevant part of the tremor band. Overall, the model supports the idea that forearm and wrist DOFs are the most critical targets for tremor suppression, but the passive device must be tuned carefully to avoid simply moving the resonance into another relevant tremor frequency range.

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Appendix

Time Responses by Input DOF

For input tremor torque in DOF 1:

Shoulder

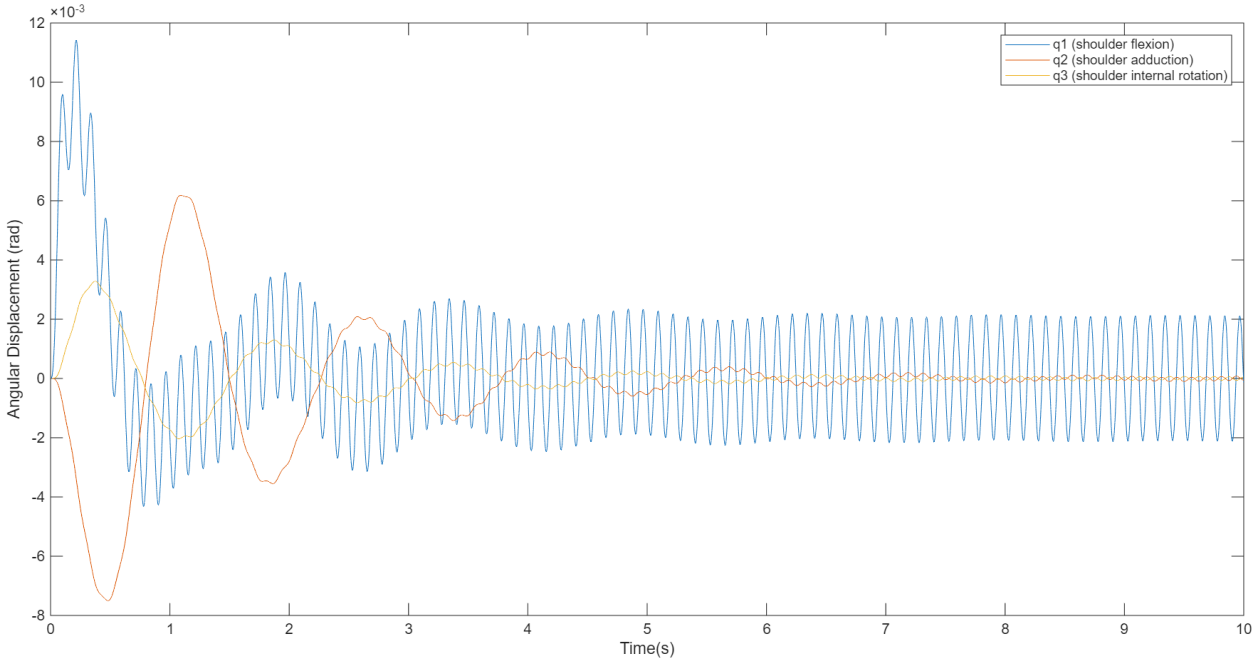


Figure 25: Time Response of Shoulder DOFs to Input Tremor Torque at DOF 1

Elbow

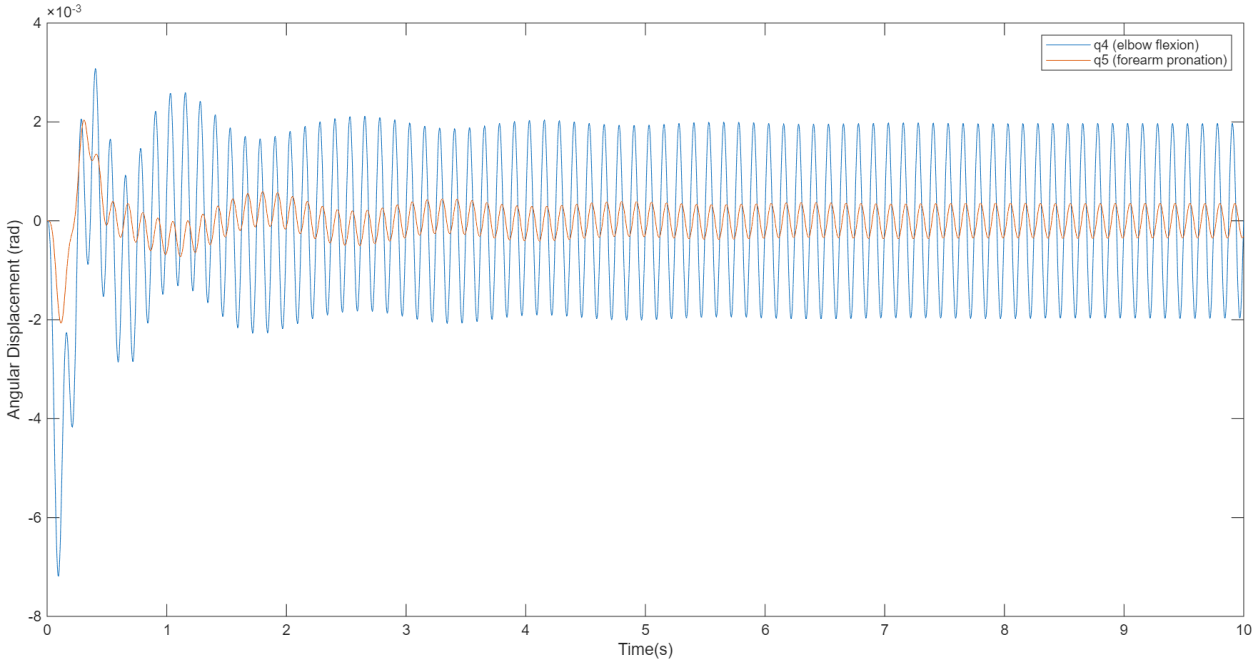


Figure 26: Time Response of Elbow DOFs to Input Tremor Torque at DOF 1

Wrist

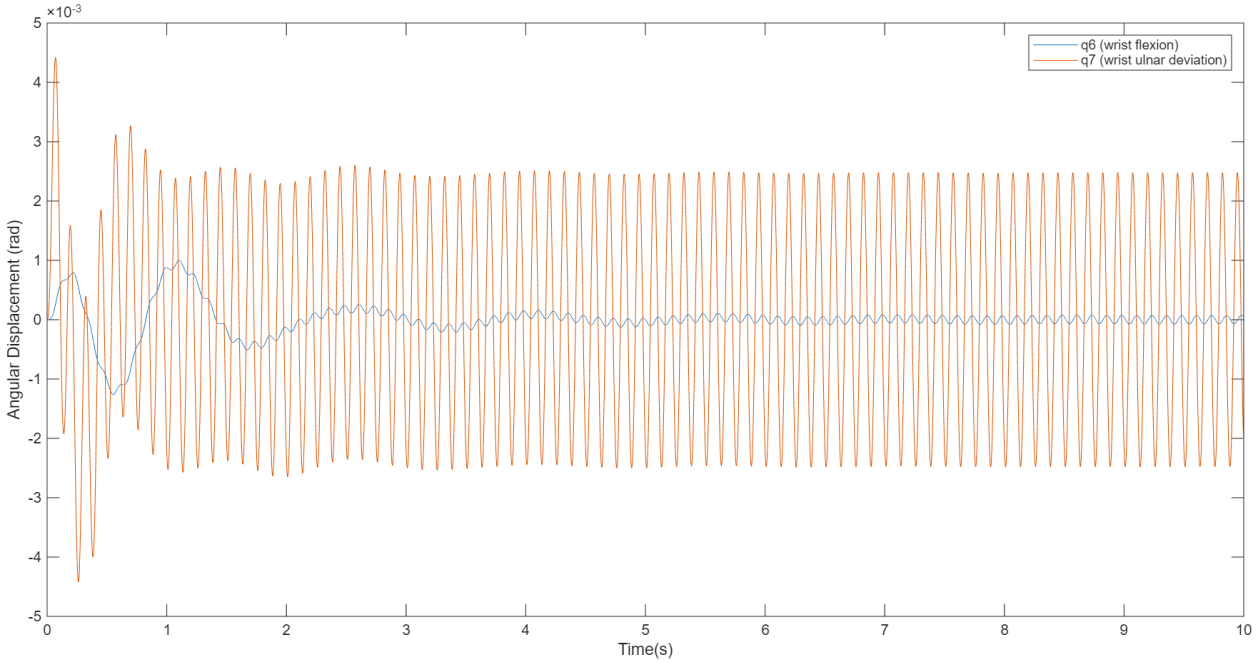


Figure 27: Time Response of Wrist DOFs to Input Tremor Torque at DOF 1

For input tremor torque in DOF 2:

Shoulder

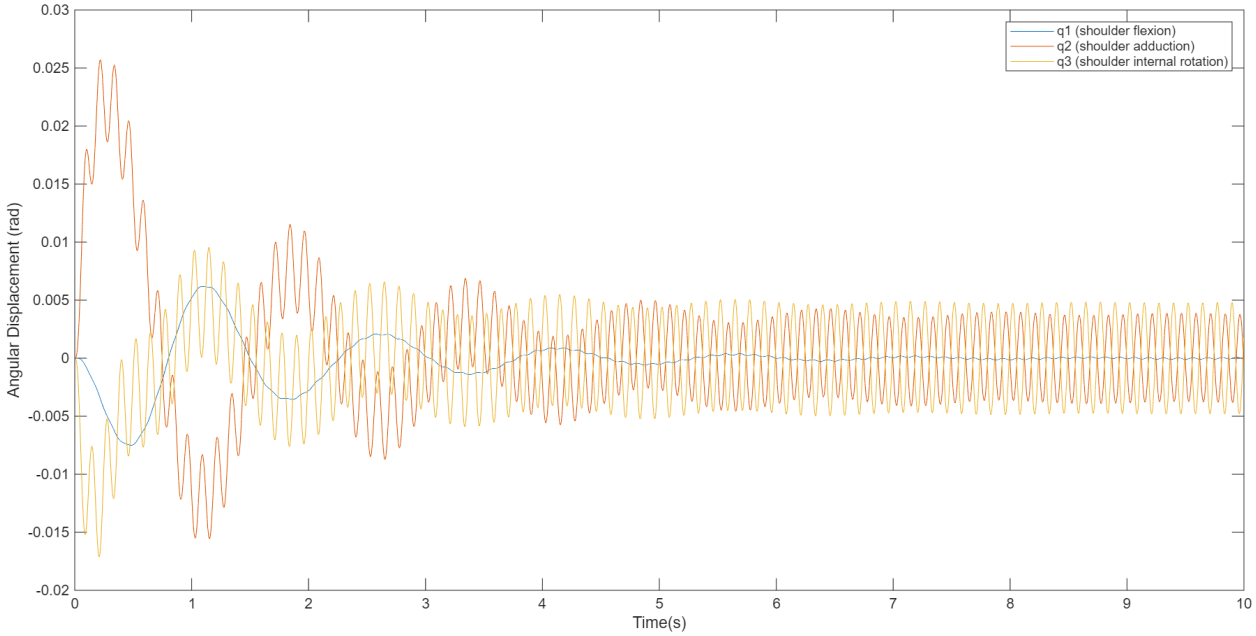


Figure 28: Time Response of Shoulder DOFs to Input Tremor Torque at DOF 2

Elbow

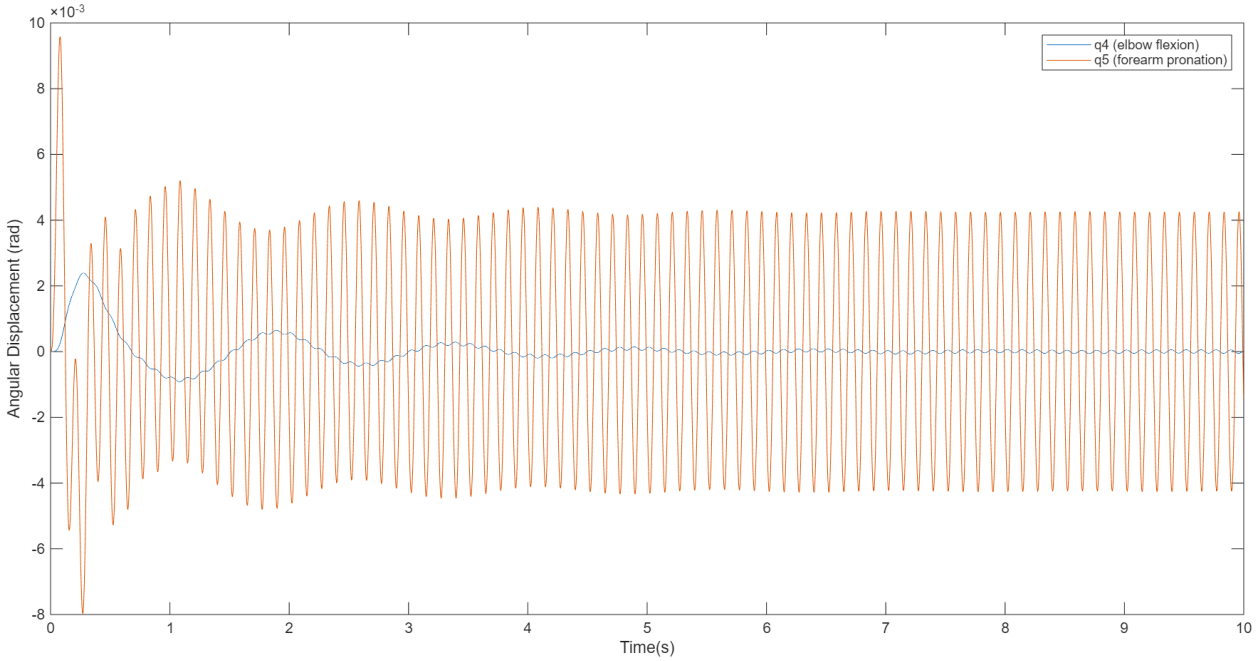


Figure 29: Time Response of Elbow DOFs to Input Tremor Torque at DOF 2

Wrist

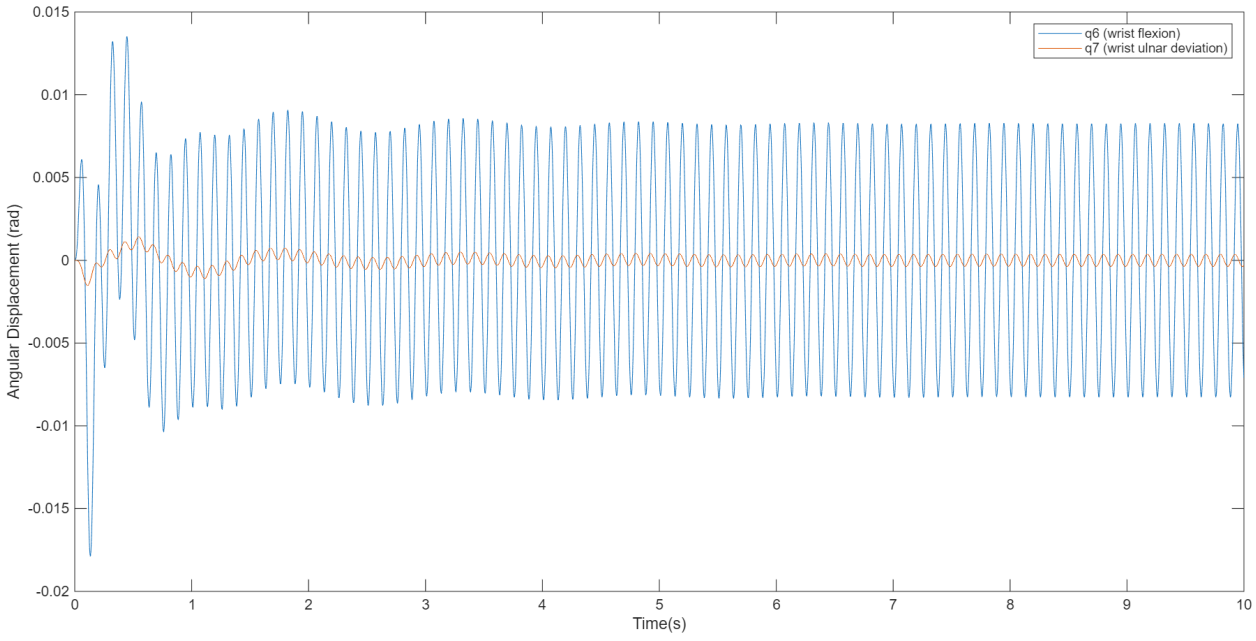


Figure 30: Time Response of Wrist DOFs to Input Tremor Torque at DOF 2

For input tremor torque in DOF 3:

Shoulder

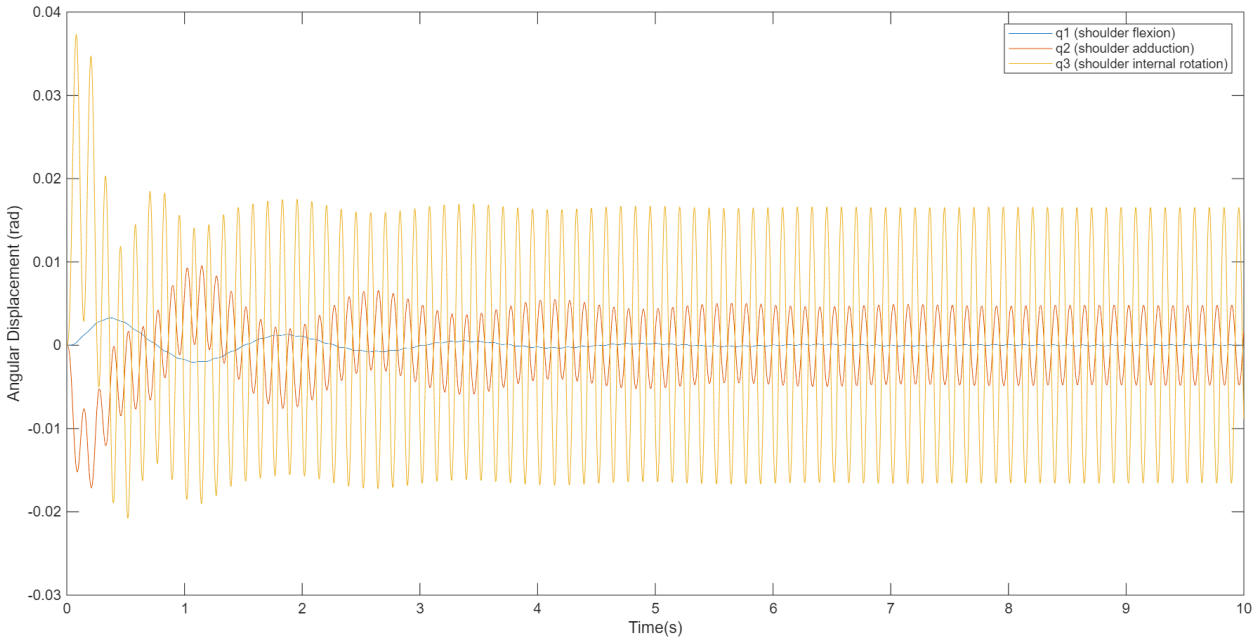


Figure 31:Time Response of Shoulder DOFs to Input Tremor Torque at DOF 3

Elbow

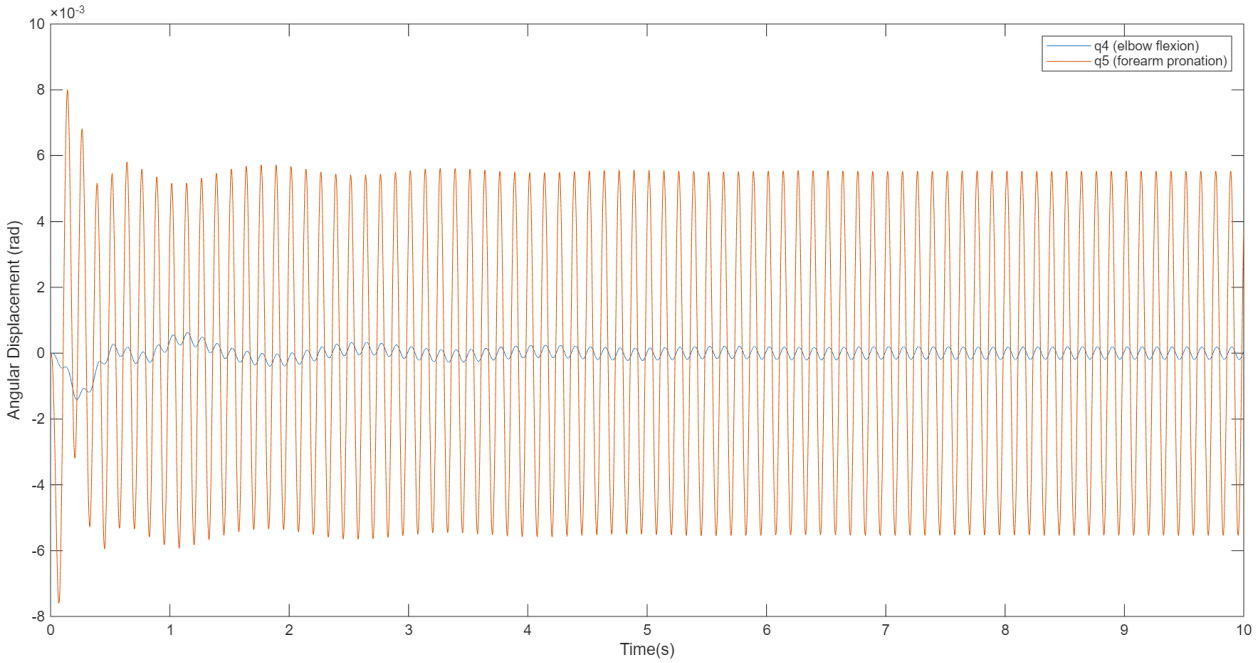


Figure 32: Time Response of Elbow DOFs to Input Tremor Torque at DOF 3

Wrist

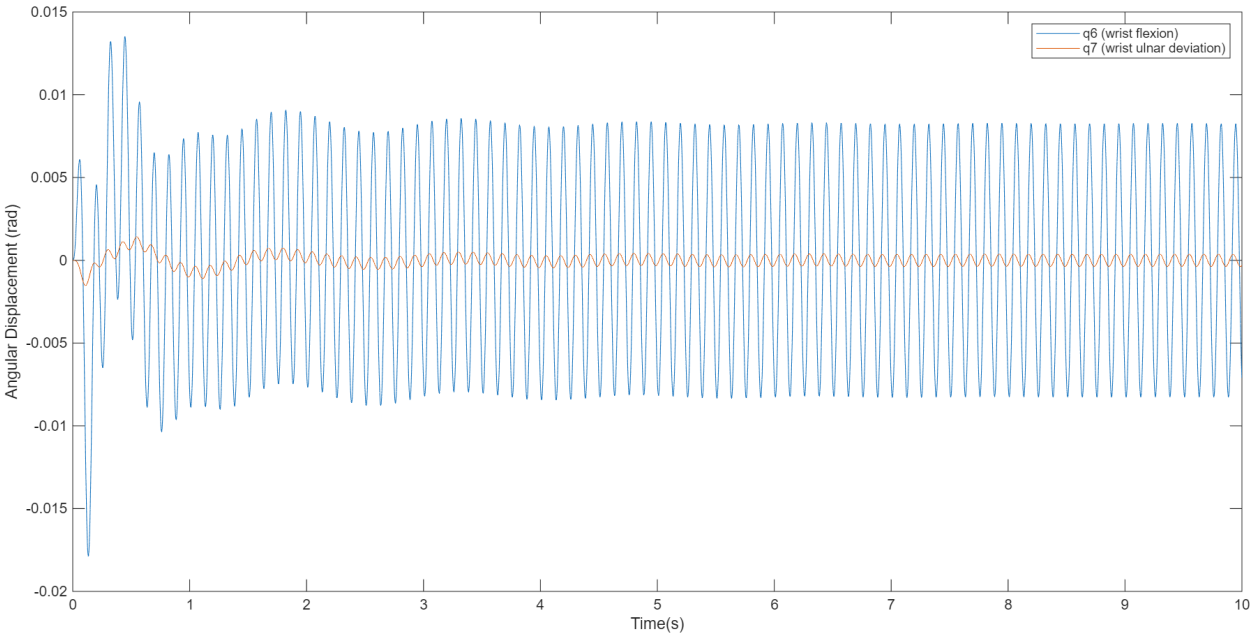


Figure 33: Time Response of Wrist DOFs to Input Tremor Torque at DOF 3

For input tremor torque in DOF 4:

Shoulder

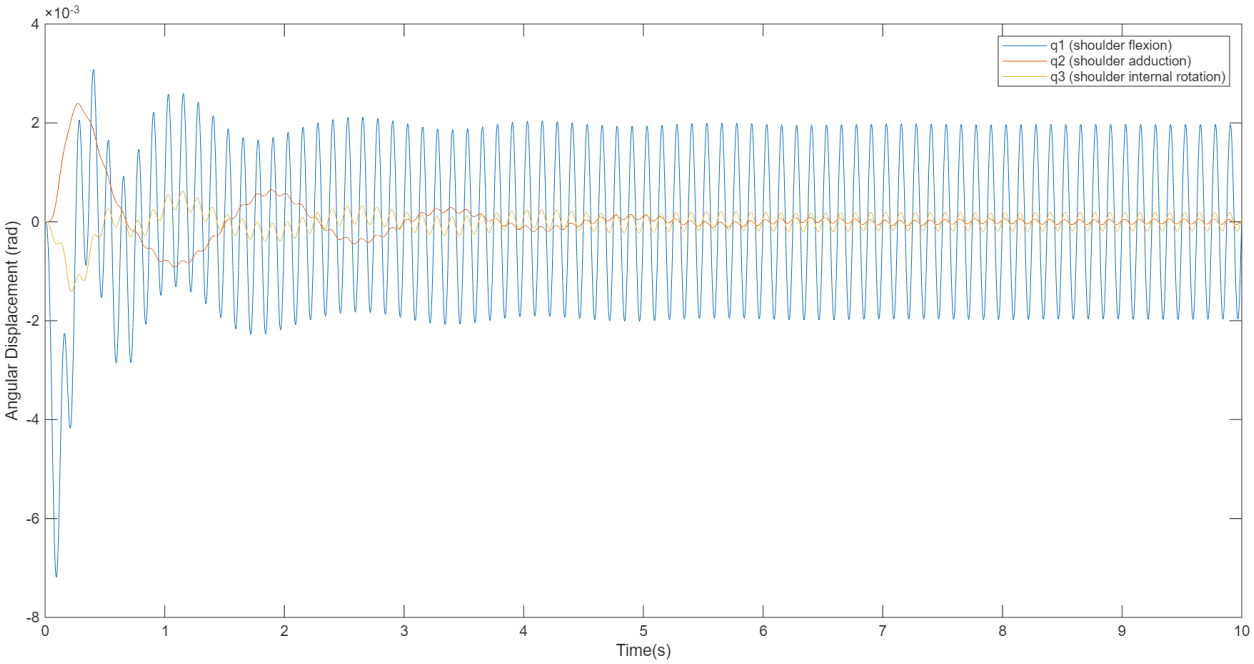


Figure 34: Time Response of Shoulder DOFs to Input Tremor Torque at DOF 4

Elbow

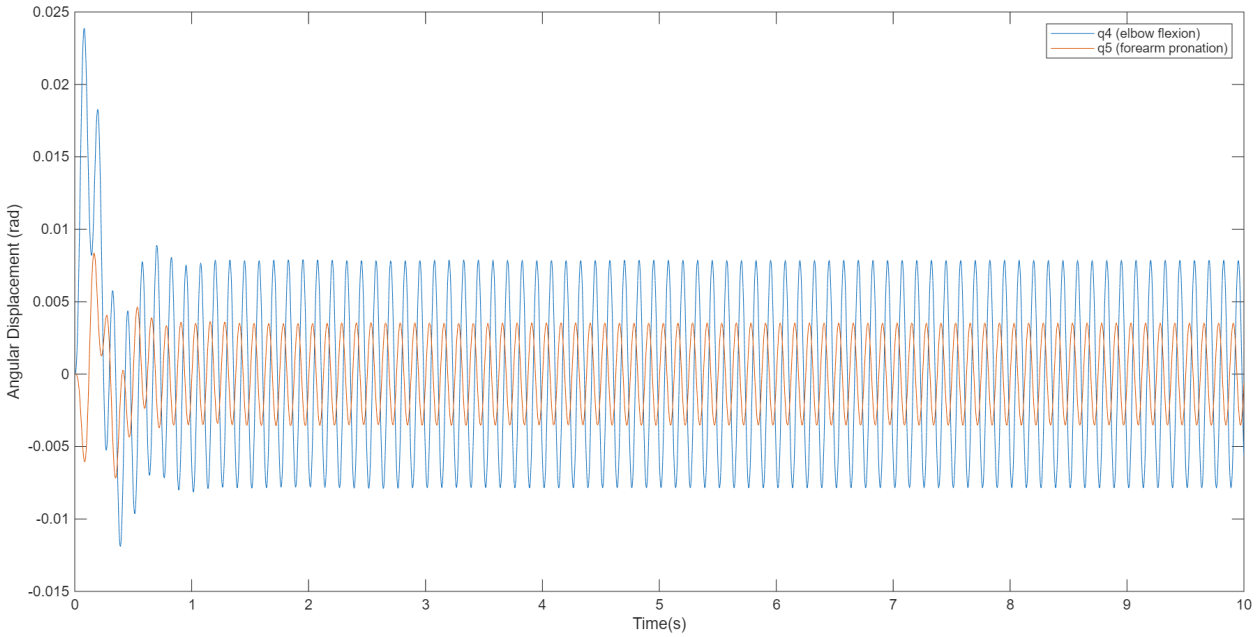


Figure 35: Time Response of Elbow DOFs to Input Tremor Torque at DOF 4

Wrist

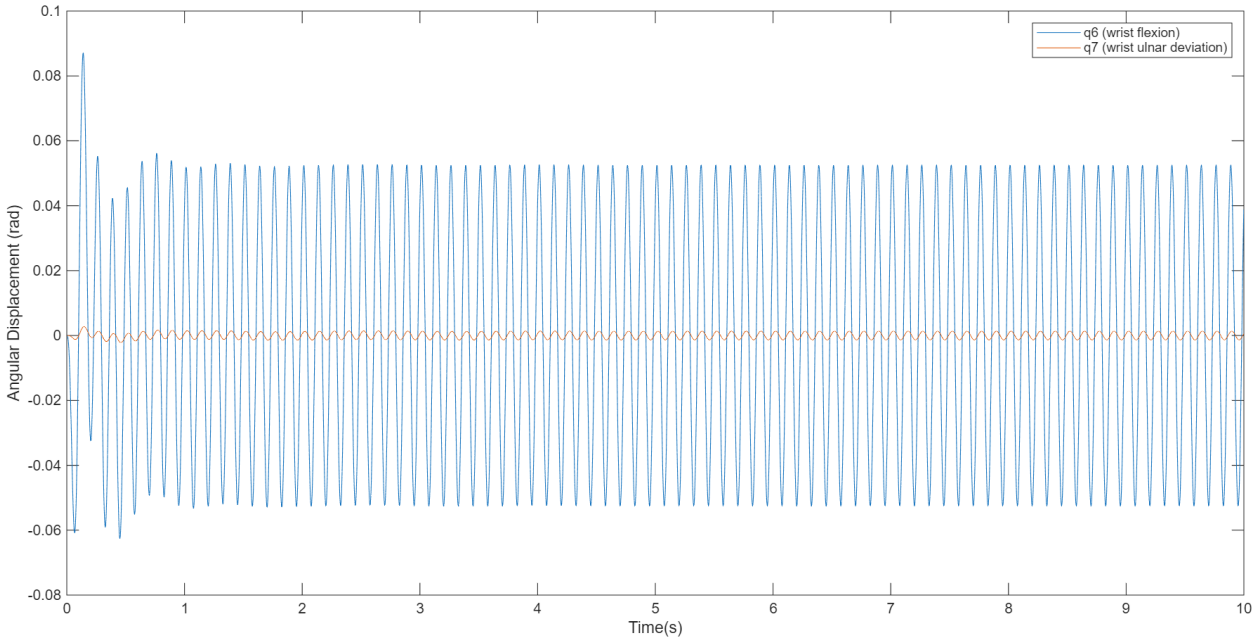


Figure 36: Time Response of Wrist DOFs to Input Tremor Torque at DOF 4

For input tremor torque in DOF 5:

Shoulder

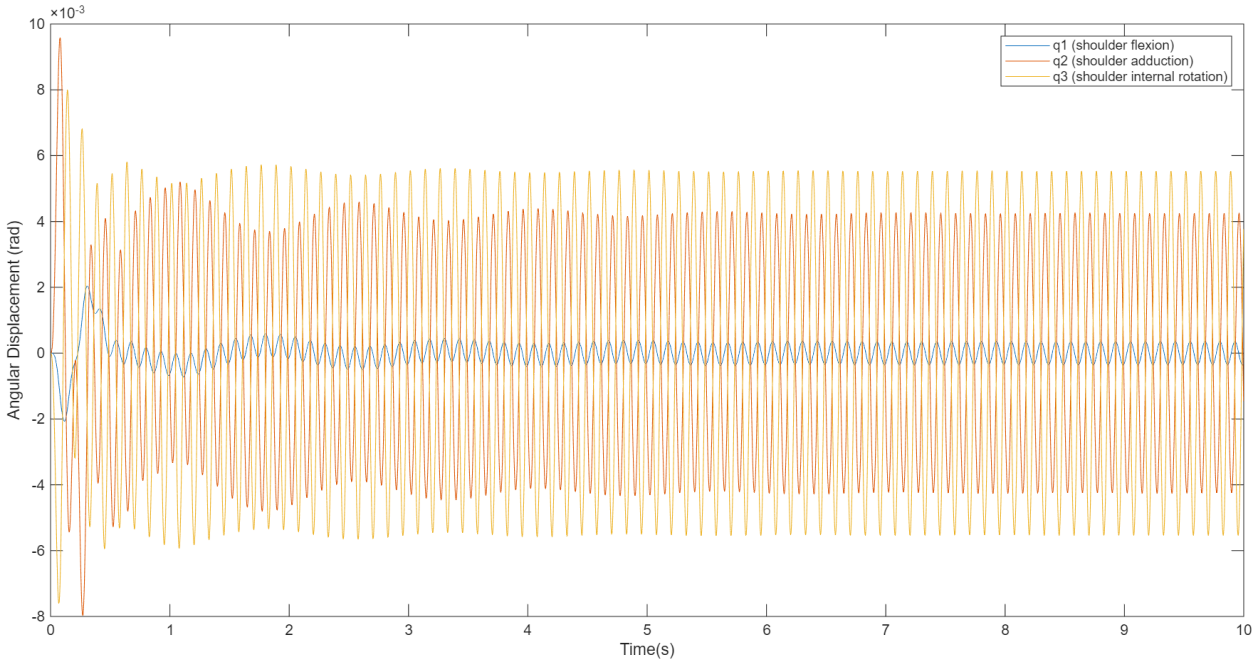


Figure 37: Time Response of Shoulder DOFs to Input Tremor Torque at DOF 5

Elbow

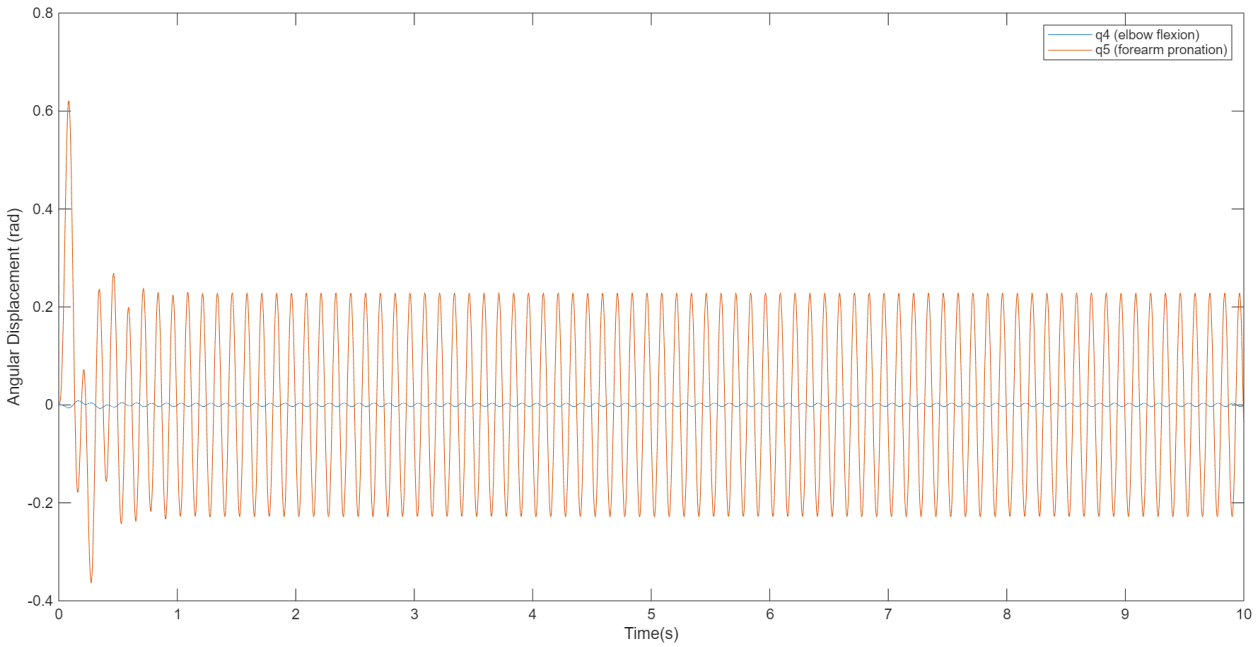


Figure 38: Time Response of Elbow DOFs to Input Tremor Torque at DOF 5

Wrist

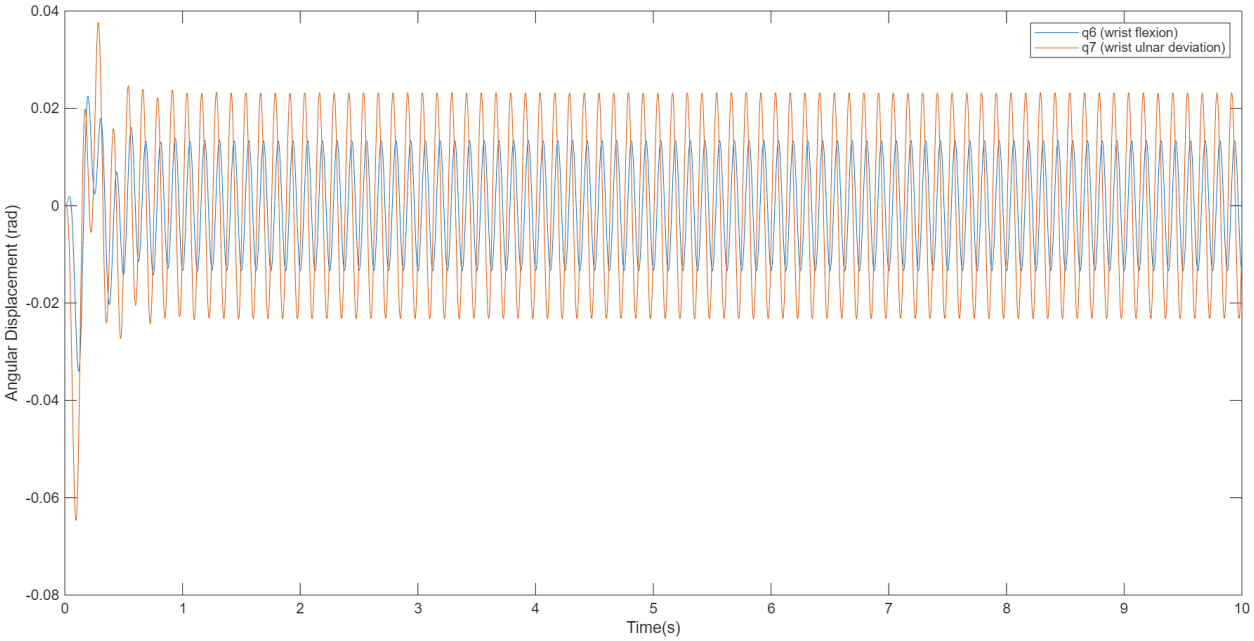


Figure 39: Time Response of Wrist DOFs to Input Tremor Torque at DOF 5

For input tremor torque in DOF 6:

Shoulder

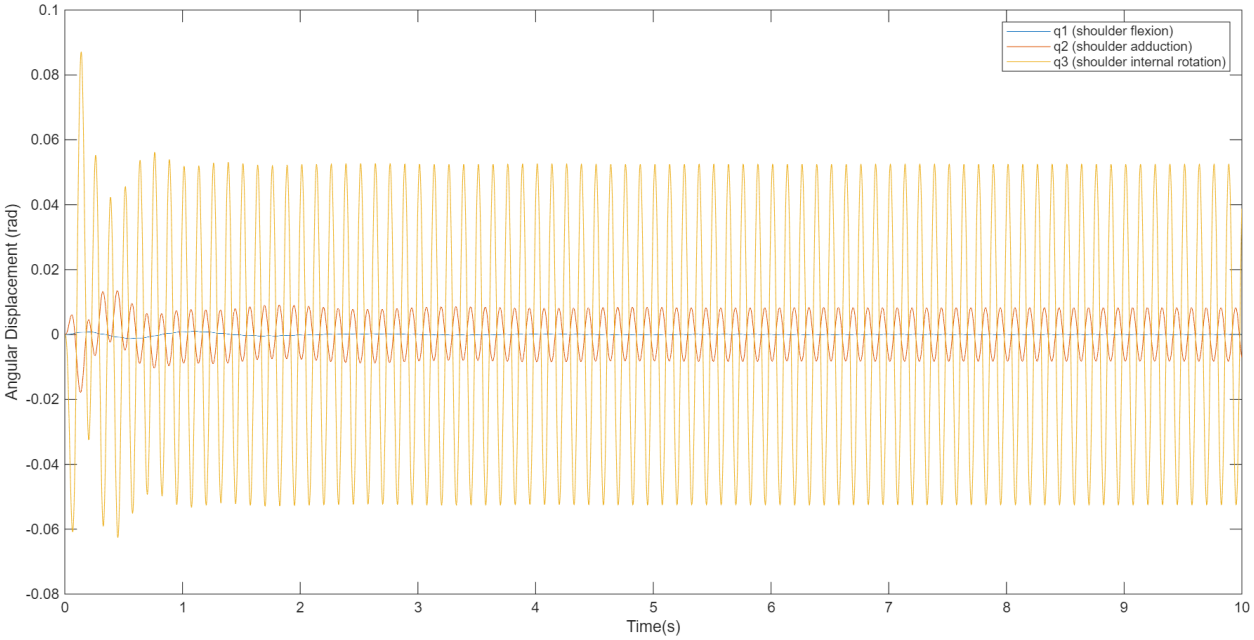


Figure 40: Time Response of Shoulder DOFs to Input Tremor Torque at DOF 6

Elbow

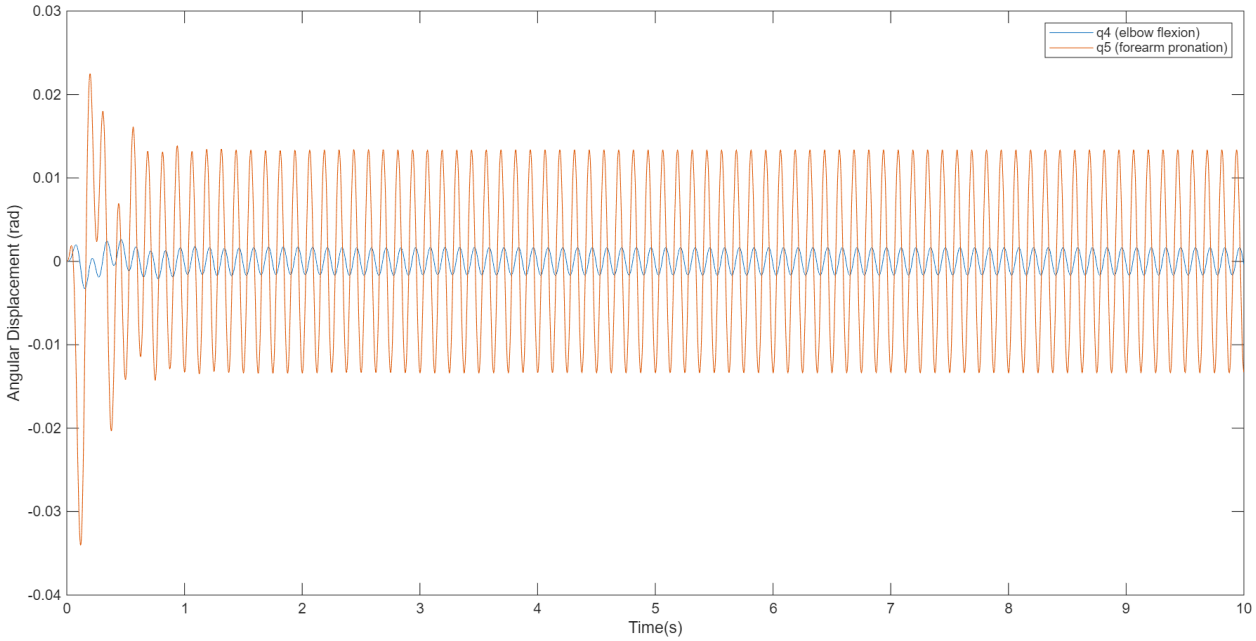


Figure 41: Time Response of Elbow DOFs to Input Tremor Torque at DOF 6

Wrist

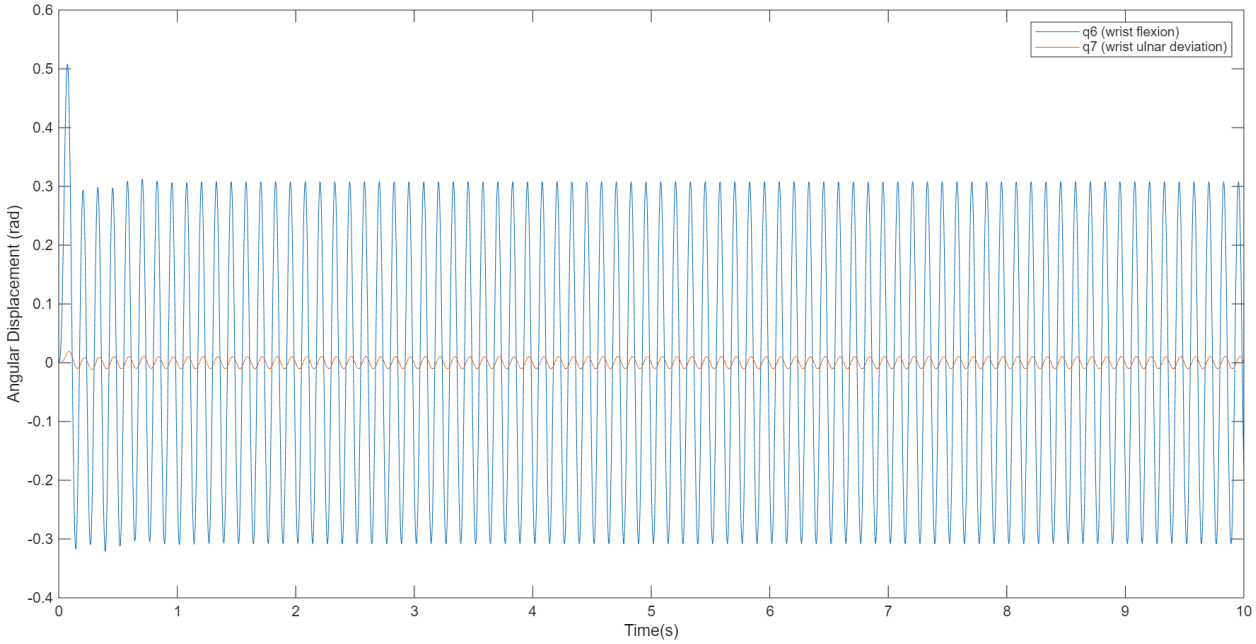


Figure 42: Time Response of Wrist DOFs to Input Tremor Torque at DOF 6

For input tremor torque in DOF 7:

Shoulder

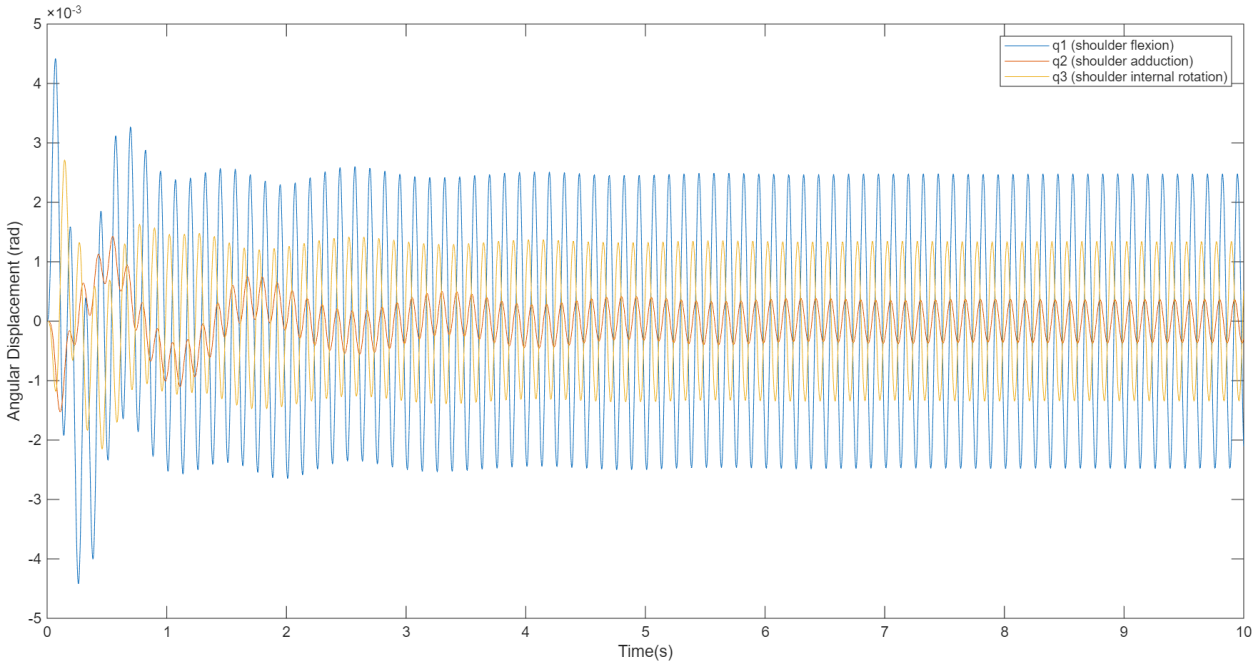


Figure 43: Time Response of Shoulder DOFs to Input Tremor Torque at DOF 7

Elbow

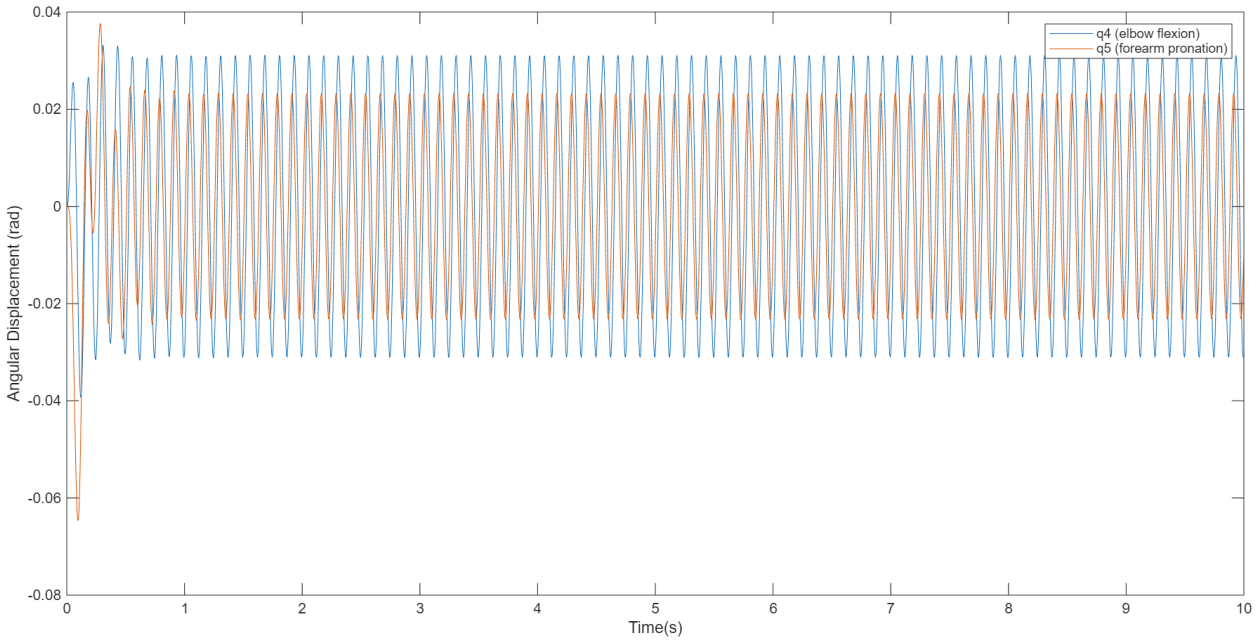


Figure 44: Time Response of Elbow DOFs to Input Tremor Torque at DOF 7

Wrist

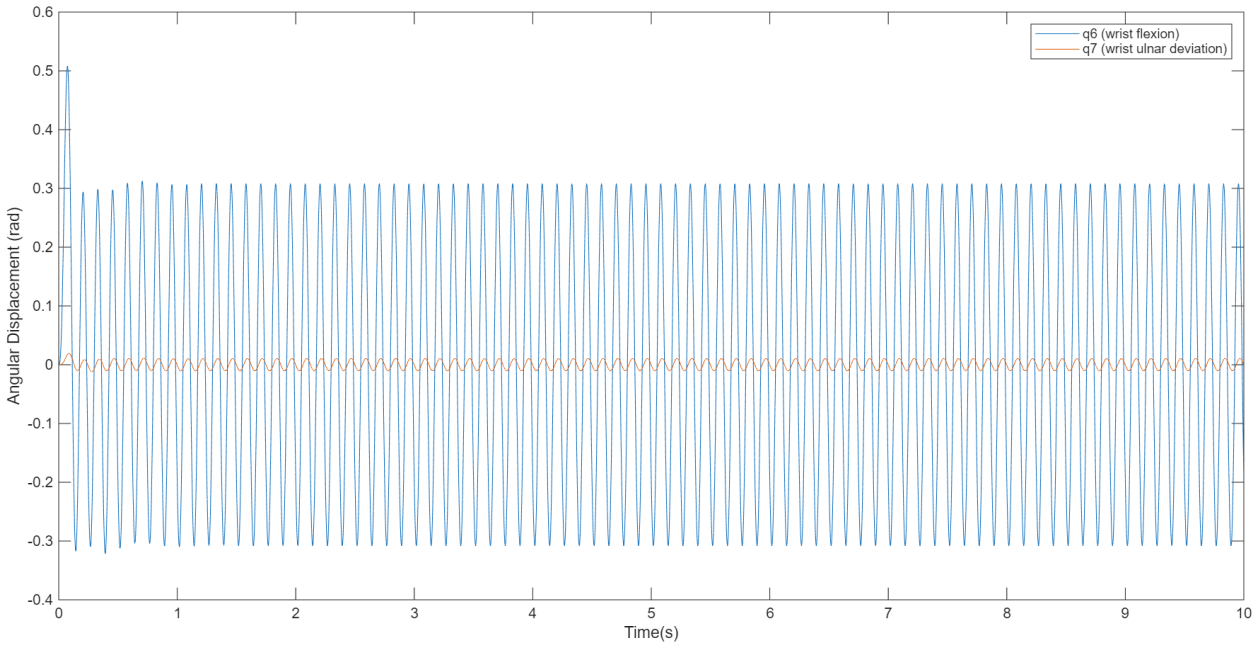


Figure 45: Time Response of Wrist DOFs to Input Tremor Torque at DOF 7

Real and Imaginary parts of FRF

Shoulder

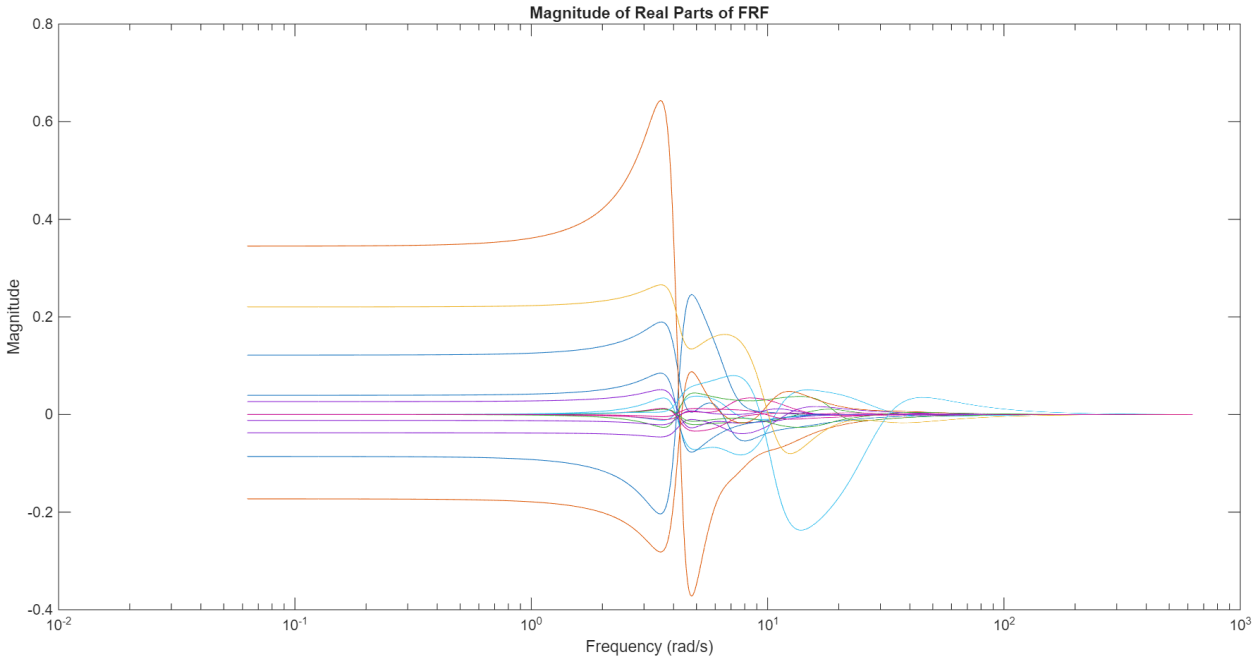


Figure 46: Magnitude of Real Parts of Shoulder FRF

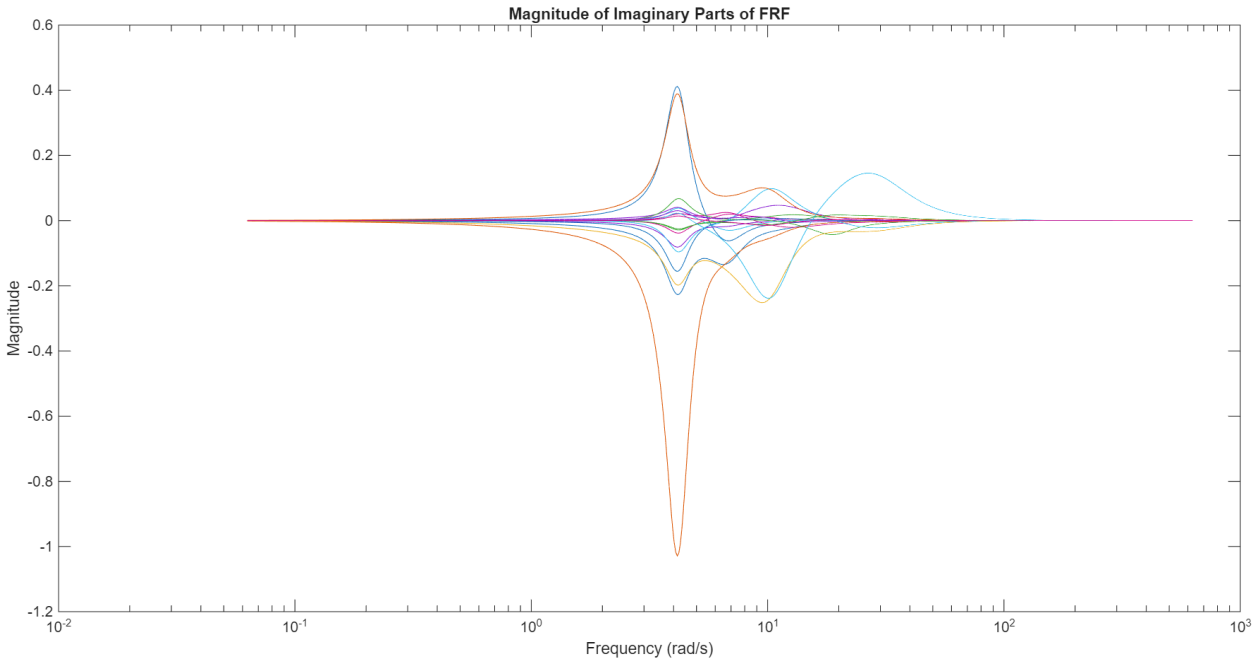


Figure 47: Magnitude of Imaginary Parts of Shoulder FRF

Elbow

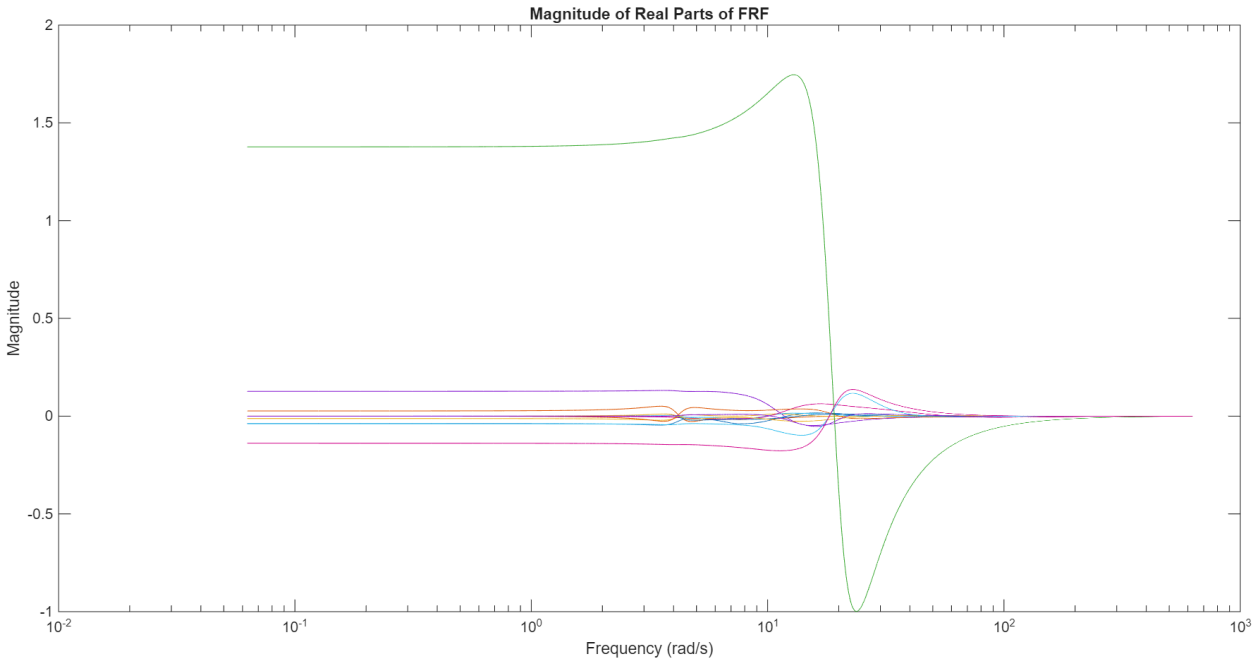


Figure 48: Magnitude of Real Parts of Elbow FRF

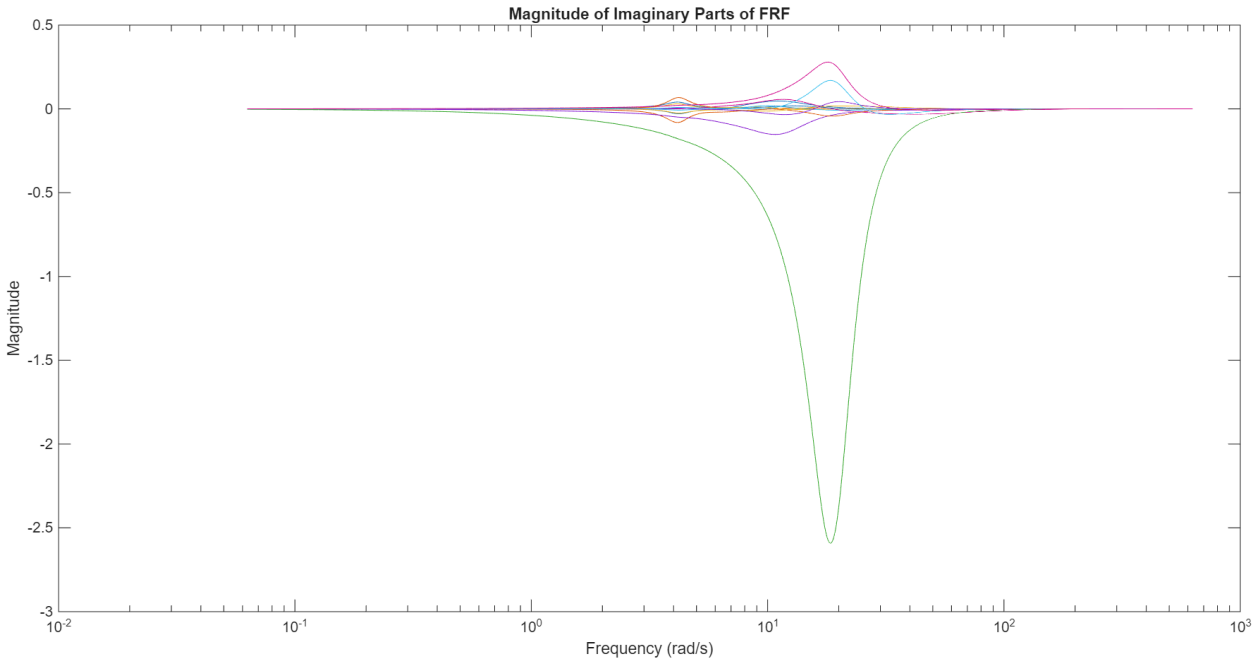


Figure 49: Magnitude of Imaginary Parts of Elbow FRF

Wrist

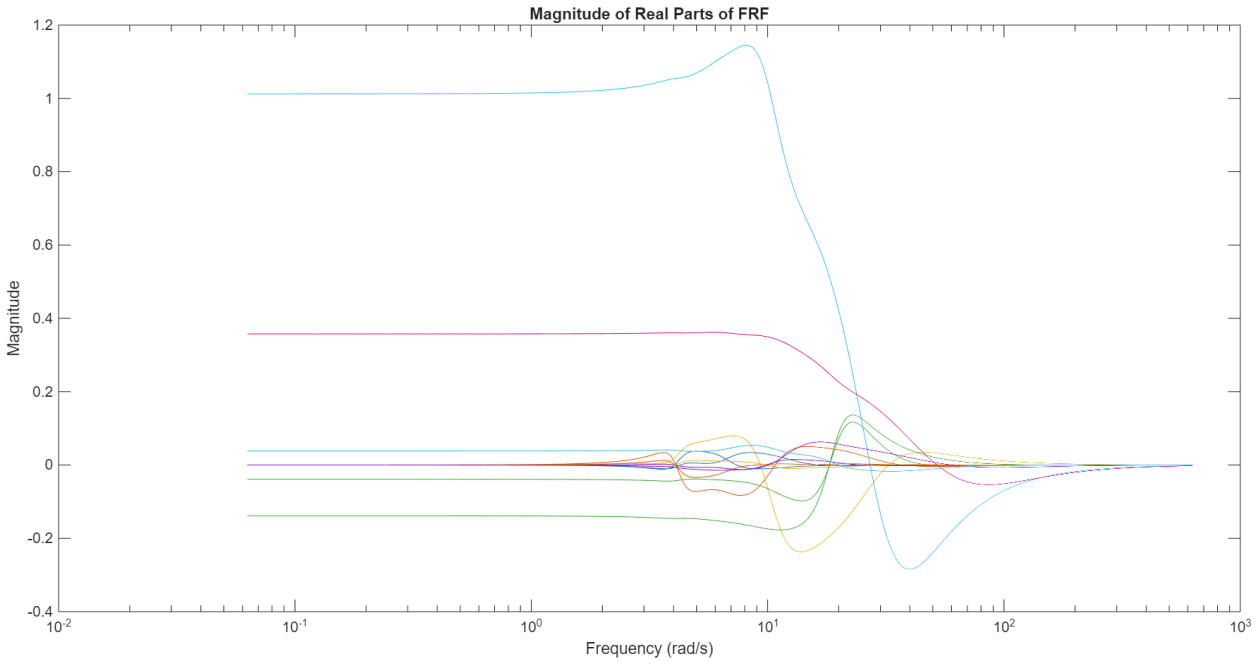


Figure 50: Magnitude of Real Parts of Wrist FRF

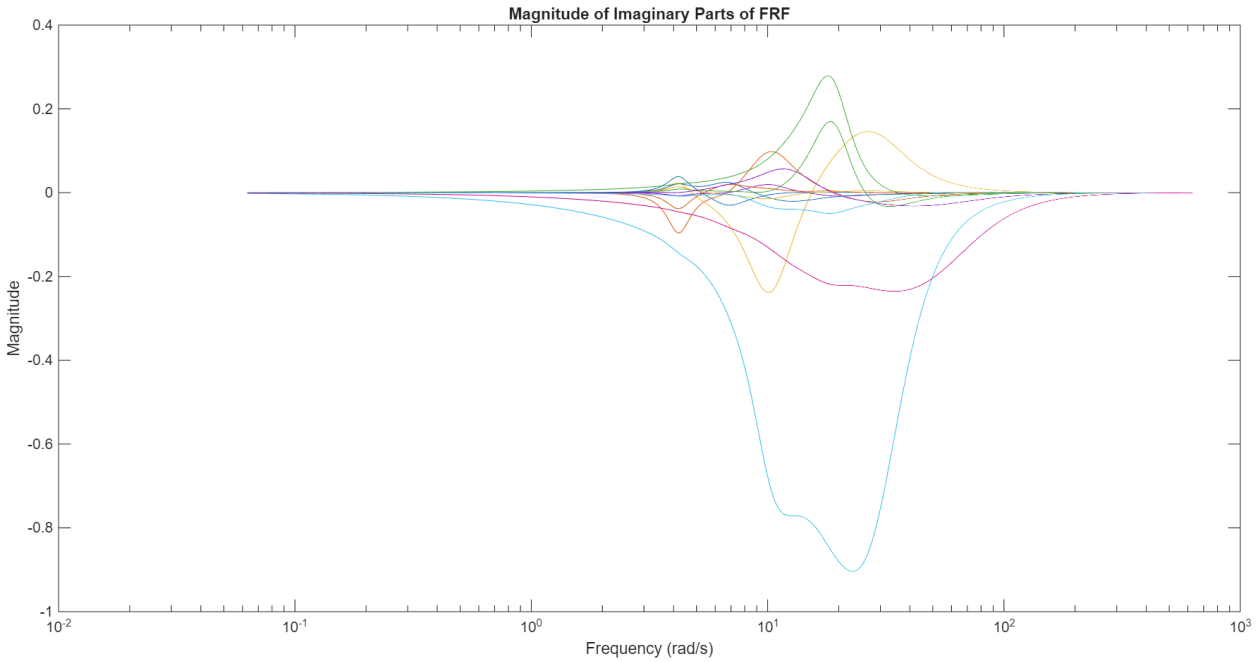


Figure 51: Magnitude of Imaginary Parts of Wrist FRF

MATLAB Code

For Baseline Tremor Behavior

```
% Ray Ramadhar
% ME 301
% 4/8/2026

clc;
clear all;
close all;

%% Defnitions
% Deifne the tremor frequencies
w_t = 4*2*pi;

% Define a Time Vector
N = 10000; % Samples
dt = 0.001; % s/Sample
t = (0:N-1).'*dt; % s

% Define a Frequency Domain
Nf = 5000;
df = 0.01;
f = (0:N-1).'*df;
w = 2*pi*f;

% Setting up the Laplace domain
s = tf('s');

% Given (input) Parameters Imedance Matrices
K =    [10.80   2.626   0.279   2.670   0       0       0;
        2.626  5.468   3.821   0       0       0       0;
        0.279  3.821   7.486   0       0       0       0;
        2.670  0       0       8.670   0       0       0;
        0      0       0       0       0.756   0.018   0.291;
        0      0       0       0       0.018   0.992  -0.099;
        0      0       0       0       0.291  -0.099   2.920];
K_imp = K/s;


R =   [0.756   0.184   0.020   0.187   0       0       0;
       0.184   0.383   0.267   0       0       0       0;
       0.020   0.267   0.524   0       0       0       0;
       0.187   0       0       0.607   0       0       0;
       0       0       0       0       0.021   0.001   0.008;
       0       0       0       0       0.001   0.028  -0.003;
       0       0       0       0       0.008  -0.003   0.082];
R_imp = R;

I =   [0.269   0       0       0.076   0       0      -0.014;
       0       0.196   0.083   0      -0.002   0.009   0;
       0       0.083   0.079   0       0       0.011   0;
       0.076   0       0       0.076   0       0      -0.012;
       0      -0.002   0       0       0.002   0       0;
       0       0.009   0.011   0       0       0.003   0;
     -0.014    0       0      -0.012   0       0       0.003];
I_imp = I*s;

%% State Space Form
E = eye(7);
F = zeros(7,7);

% A matrix:
A = [F, E;
```

```
    -(I \ K), -(I \ R)];

% B matrix:
B = [F;
     inv(I)]; % Or I \ eye(7)

% C matrix:
C = [E, F];

% D matrix:
D = F;

sys_ss = ss(A,B,C,D);

% Checking poles of the system to see if its stable
figure(1)
p = pole(sys_ss);
plot(real(p), imag(p), 'x', 'MarkerSize', 10, 'LineWidth', 2);
title('System Poles');

%% Trying to eliminate and potential instability we sw before with min real
sys_dense = full(sys_ss);
sys_ss = ss(sys_dense);

tol = 1e-4;
sys_clean = minreal(sys_ss, tol);

[sys_balanced,~] = balreal(sys_clean);

%% Trying to plot the angular position
u = zeros(length(t), 7);
u(:, 6) = 1/3 * sin(w_t * t);


[q, t] = lsim(sys_balanced,u,t);

figure(2)
% Plotting All
plot(t,q)
legend('q1 (shoulder flexion)', 'q2 (shoulder adduction)','q3 (shoulder internal rotation)','q4
(elbow flexion)','q5 (forearm pronation)','q6 (wrist flexion)','q7 (wrist ulnar deviation)')
% Plotting Shoulder
% plot(t,q(:,1),t,q(:,2),t,q(:,3))
% legend('q1 (shoulder flexion)', 'q2 (shoulder adduction)','q3 (shoulder internal rotation)')
% Plotting Elbow/Forearm
% plot(t,q(:,4),t,q(:,5))
% legend('q4 (elbow flexion)', 'q5 (forearm pronation)')
% Plotting Wrist
% plot(t,q(:,6),t,q(:,7))
% legend('q6 (wrist flexion)', 'q7 (wrist ulnar deviation)')
% xlabel('Time(s)');
% ylabel('Angular Displacement (rad)')

%% Extracting the natural frequencies, damping ratios, and resonant frequencies
[wn_both, zeta_both] = damp(sys_balanced);
wns = unique(wn_both);
zetas = unique(zeta_both);

% Calculating the resonant freuqnecies at each pole
w_r = wns .* sqrt(1 - 2 * zetas.^2);

```

```
% Removing imaginary results (where zeta > 0.707)
w_r(zetas > 0.707) = 0;

%% Defining H and plotting
H = freqresp(squeeze(sys_balanced), w);
n = size(H,1);
m = size(H,2);
rowsToPlot = 4;
numRows = length(rowsToPlot);

% Maginitude of H
figure(3)
% for idx = 1:numRows
%     i = rowsToPlot(idx);
%     for j = 1:m
%         subplot(numRows, m, (idx-1)*m + j);
%         semilogx(w, squeeze(20*log10(abs((H(i,j,:))))));
%         grid on;
%         title(['From: In(' num2str(j) ') to Out(' num2str(i) ')'], 'FontSize', 9);
%         if idx == numRows, xlabel('Frequency (rad/s)'); end
%         if j == 1, ylabel('Mag (dB)'); end
%     end
% end
for i = 1:n;
    for idx = 1:numRows
        j = rowsToPlot(idx);
        semilogx(w, squeeze(abs((H(i,j,:)))));
        hold on;
        grid on;
        title(['From: In(Tremor Torque in Elbow Flexion DOF) To: Out(All DOFS)'], 'FontSize', 9);
        if idx == numRows, xlabel('Frequency (rad/s)'); end
        if j == 1, ylabel('Mag (rad/N*m)'); end
    end
    legend('q1 (shoulder flexion)', 'q2 (shoulder abduction)','q3 (shoulder internal
rotation)','q4 (elbow flexion)','q5 (forearm pronation)','q6 (wrist flexion)','q7 (wrist ulnar
deviation)')
end
grid on;
xlabel('Frequency (rad/s)');
ylabel('Magnitude (rad/N*m)');
hold off;

% Imaginary of the FRF matrix as functions of frequency
figure(4)
for i = 1:7
        for j = 1:m
            Himag = squeeze(imag(H(i,j,:)));
            semilogx(w, Himag);
            hold on;
        end
end
hold off;

xlabel('Frequency (rad/s)');
ylabel('Magnitude');
title('Magnitude of Imaginary Parts of FRF');

% Real of the FRF matrix as functions of frequency
figure(5)
for i = 1:7
        for j = 1:m
```

```
            Hreal = squeeze(real(H(i,j,:)));
            semilogx(w, Hreal);
            hold on;
        end
end
hold off;

xlabel('Frequency (rad/s)');
ylabel('Magnitude');
title('Magnitude of Real Parts of FRF');

% %% Defining the Modes of the system
% % Reshaping the 7x7x(at # of w's) to 49x(at # of w's)
% im = reshape(imag(H), [n*m, length(imag(H))]).';
% wns_idx = zeros(size(wns));
%
% % Finding each index
% for g = 1:length(wns)
%     [~, wns_idx(g)] = min(abs(w - wns(g)));
% end
%
% % Mode Shape Matrix (7x7)
% A = im(wns_idx,1:n);
%
% % Finding and plotting the mode shapes of the system
% % May need a scale factor
% A = [zeros(1, size(A, 2)); A];
% Mode1 = A(:,1);
% Mode2 = A(:,2);
% Mode3 = A(:,3);
% Mode4 = A(:,4);
% Mode5 = A(:,5);
% Mode6 = A(:,6);
% Mode7 = A(:,7);
%
% % Defining Axes
% figure(6)
% y = 0:7;
% ylim([0,7.75])
% yticks(y)
% yticklabels({'q1','q2','q3','q4','q5','q6','q7'});
% ylabel('Angular Coordinate')
% x = 1:7;
% xlim([0,7])
% xticks([0, 1, 2, 3, 4, 5, 6, 7]);
% xticklabels([0, 1, 2, 3, 4, 5, 6, 7]);
% xlabel('Mode Shape')
%
% % Space the modes 1 apart from each other
% plot(Mode1+1,y)
% hold on;
% plot(Mode2+2,y)
% hold on;
% plot(Mode3+3,y)
% hold on;
% plot(Mode4+4,y)
% hold on;
% plot(Mode5+5,y)
% hold on;
% plot(Mode6+6,y)
% hold on;
```

```
% plot(Mode7+7,y)
% hold on;
%
% % Adding zero lines
% xline(x, '--k')
% text(1.05,6.8,'0')
% text(2.05,6.8,'0')
% text(3.05,6.8,'0')
% text(4.05,6.8,'0')
% text(5.05,6.8,'0')
% text(6.05,6.8,'0')
% text(7.05,6.8,'0')


%% Unnormalized mode-shape plot
% Uses raw eigenvectors directly from K*Phi = I*Phi*Lambda
M = I;   % Your inertia/mass matrix
C = R;   % Your damping matrix

% K*Phi = M*Phi*Lambda
[Phi, Lambda] = eig(K, M);
% Extract eigenvalues and remove tiny numerical imaginary parts
lambda = real(diag(Lambda));
Phi = real(Phi);
% Natural frequencies
wn = sqrt(lambda);      % rad/s
fn = wn/(2*pi);         % Hz
% Sort modes from lowest to highest natural frequency
[wn, sort_idx] = sort(wn);
fn = fn(sort_idx);
lambda = lambda(sort_idx);
Phi = Phi(:, sort_idx);
% Raw eigenvectors from eig(K,M)
Phi_raw = Phi;
% Display natural frequencies and raw mode shape matrix
disp('Natural frequencies from unnormalized eigenvectors:')
freq_table = table((1:7).', lambda, wn, fn, ...
    'VariableNames', {'Mode','Eigenvalue_lambda','NaturalFreq_rad_s','NaturalFreq_Hz'});
disp(freq_table)
disp('Raw unnormalized mode shape matrix Phi:')
disp(Phi_raw)
coord_labels = {'q1 Shoulder Flexion', ...
                'q2 Shoulder Adduction', ...
                'q3 Shoulder Internal Rotation', ...
                'q4 Elbow Flexion', ...
                'q5 Forearm Pronation', ...
                'q6 Wrist Flexion', ...
                'q7 Wrist Ulnar Deviation'};
% Plot all unnormalized modes on one spaced plot
figure(21);
hold on; grid on;
y = 1:7;
visual_scale = 1;
for mode = 1:7
    x_offset = mode;
    x_mode = x_offset + visual_scale * Phi_raw(:,mode);

    plot(x_mode, y, '-o', 'LineWidth', 2, 'MarkerFaceColor', 'auto')
    xline(x_offset, '--k', 'LineWidth', 1)
end
```

```
set(gca, 'YDir', 'reverse')
yticks(1:7)
yticklabels(coord_labels)
xlabel('Mode number with raw mode-shape deformation')
ylabel('Angular coordinate')
title('Unnormalized 7-DOF Arm Mode Shapes')
xlim([0.4 7.6])
ylim([0.5 7.5])


```

For No Suppression vs. Suppression comparison

```
clc;
clear all;
close all;

%% Defnitions
% Deifne the tremor frequencies
w_t = 4*2*pi;

% Define a Time Vector
N = 10000; % Samples
dt = 0.001; % s/Sample
t = (0:N-1).'*dt; % s

% Define a Frequency Domain
Nf = 5000;
df = 0.01;
f = (0:N-1).'*df;
w = 2*pi*f;

% Setting up the Laplace domain
s = tf('s');

% Given (input) Parameters Imedance Matrices
K_no_sup =    [10.80   2.626   0.279   2.670   0       0       0;
       2.626  5.468   3.821   0       0       0       0;
       0.279  3.821   7.486   0       0       0       0;
       2.670  0       0       8.670   0       0       0;
       0      0       0       0       0.756   0.018   0.291;
       0      0       0       0       0.018   0.992  -0.099;
       0      0       0       0       0.291  -0.099   2.920];
K_device = [0   0   0   0   0       0       0;
       0  0   0   0       0       0       0;
       0  0   0   0       0       0       0;
       0  0       0       0   0       0       0;
       0      0       0       0       0   0   0;
       0      0       0       0       0   0  0;
       0      0       0       0       0  0   0];
K_with_sup = K_no_sup + K_device;
% K_imp = K/s;


R =   [0.756   0.184   0.020   0.187   0       0       0;
      0.184   0.383   0.267   0       0       0       0;
      0.020   0.267   0.524   0       0       0       0;
      0.187   0       0       0.607   0       0       0;
      0       0       0       0       0.021   0.001   0.008;
      0       0       0       0       0.001   0.028  -0.003;
      0       0       0       0       0.008  -0.003   0.082];
R_imp = R;

I_no_sup =   [0.269   0       0       0.076   0       0      -0.014;
      0       0.196   0.083   0      -0.002   0.009   0;
      0       0.083   0.079   0       0       0.011   0;
      0.076   0       0       0.076   0       0      -0.012;
      0      -0.002   0       0       0.002   0       0;
      0       0.009   0.011   0       0       0.003   0;
     -0.014   0       0      -0.012   0       0       0.003];
```

```
I_device = [
    0.0468    0    0    0         0         0         0.004446;
    0         0.0468    0         0         0         0         0;
    0    0    0.0468    0         0         0.004446  0;
    0         0    0         0.0234    0         0         0.0039;
    0         0    0         0         0.000416  0         0;
    0         0    0.004446  0         0         0.0234    0;
    0.004446  0    0         0.0039    0         0         0.00065
];
I_with_sup = I_no_sup + I_device;
% I_imp = I*s;

%% State Space Form (No Supression)
E = eye(7);
F = zeros(7,7);

% A matrix:
A = [F, E;
    -(I_no_sup \ K_no_sup), -(I_no_sup \ R)];

% B matrix:
B = [F;
     inv(I_no_sup)]; % Or I \ eye(7)

% C matrix:
C = [E, F];

% D matrix:
D = F;

sys_ss_no_sup = ss(A,B,C,D);

% Checking poles of the system to see if its stable
% figure(1)
% p = pole(sys_ss);
% plot(real(p), imag(p), 'x', 'MarkerSize', 10, 'LineWidth', 2);
% title('System Poles');

%% State Space Form (With Supression)
E = eye(7);
F = zeros(7,7);

% A matrix:
A = [F, E;
    -(I_with_sup \ K_with_sup), -(I_with_sup \ R)];

% B matrix:
B = [F;
     inv(I_with_sup)]; % Or I \ eye(7)

% C matrix:
C = [E, F];

% D matrix:
D = F;

sys_ss_with_sup = ss(A,B,C,D);

% Checking poles of the system to see if its stable
% figure(1)
% p = pole(sys_ss);
```

```
% plot(real(p), imag(p), 'x'); grid on;
% title('System Poles');

%% Trying to eliminate and potential instability we sw before with min real
sys_dense_no_sup = full(sys_ss_no_sup);
sys_ss_no_sup = ss(sys_dense_no_sup);

tol = 1e-4;
sys_clean_no_sup = minreal(sys_ss_no_sup, tol);

[sys_balanced_no_sup,~] = balreal(sys_clean_no_sup);

%% Trying to eliminate and potential instability we sw before with min real
sys_dense_with_sup = full(sys_ss_with_sup);
sys_ss_with_sup = ss(sys_dense_with_sup);

tol = 1e-4;
sys_clean_with_sup = minreal(sys_ss_with_sup, tol);

[sys_balanced_with_sup,~] = balreal(sys_clean_with_sup);



%% Defining H and plotting
H_no_sup = freqresp(squeeze(sys_balanced_no_sup), w);
H_with_sup = freqresp(squeeze(sys_balanced_with_sup), w);
n = size(H_no_sup,1);
m = size(H_no_sup,2);
rowsToPlot = 5:7;
numRows = length(rowsToPlot);

% Maginitude of H
figure(3)
for idx = 1:numRows
    i = rowsToPlot(idx);
    for j = 1:m
        subplot(numRows, m, (idx-1)*m + j);
        semilogx(w, squeeze(abs((H_no_sup(i,j,:)))));
        hold on;
        semilogx(w, squeeze(abs((H_with_sup(i,j,:)))));
        grid on;
        title(['From: In(' num2str(j) ') to Out(' num2str(i) ')'], 'FontSize', 9);
        if idx == numRows, xlabel('Frequency (rad/s)'); end
        if j == 1, ylabel('Mag (rad/Nm)'); end
    end
end
grid on;
xlabel('Frequency (rad/s)');
ylabel('Magnitude (rad/N*m)');
legend('No Suppression','With Supression')
hold off;


```